

BeneFusion SP5

(This manual is also applicable to BeneFusion SP5 Ex, BeneFusion SP5 Neo,
BeneFusion SP5 TIVA Syringe Pump)

Syringe Pump

Service Manual

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- all installation operations, expansions, changes, modifications and repairs of this product are conducted by Mindray Scientific authorized personnel;
- the electrical installation of the relevant room complies with the applicable national and local requirements;
- the product is used in accordance with the instructions for use.

Preface

Manual Purpose

This manual provides detailed information about the hardware composition, assembling, disassembling, testing and troubleshooting of the equipment to support effective troubleshooting and repair. It is not intended to be a comprehensive, in-depth explanation of the product architecture or technical implementation. If you experience problems that cannot be solved, please contact our After-sales Service Department.

The introductions to this product in this Service Manual are based on the most complete configuration, so some of them may not be applicable to the product you're maintaining. In case of any question, please contact our After-sales Service Department.

Observance of the manual is a prerequisite for proper equipment maintenance and prevents equipment damage and personnel injury.

Intended Audience

This manual is for professional biomedical engineers, authorized technicians or service representatives responsible for troubleshooting, repairing and maintaining the equipment.

Revision History

This manual has a revision number. This revision number changes whenever the manual is updated due to software or technical specification change. Contents of this manual are subject to change without prior notice.

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FOR YOUR NOTES

1 Safety

1.1 Safety Information

The safety statements presented in this chapter refer to basic safety information that the operator must pay attention to and abide by when using the equipment. There are additional safety statements in other chapters or sections, which may be the same as or similar to the following, or specific to particular operations.

Dangers

- Indicates an imminent hazard that, if not avoided, could result in death, serious injury or damage to product/property.
-

WARNING

- Indicates a potential hazard or unsafe practice that, if not avoided, could result in death, serious injury or damage to product/property.
-

CAUTION

- Indicates a potential hazard or unsafe practice that, if not avoided, could result in minor personal injury, product malfunction or damage to product/property.
-

NOTE

- Provides application tips or other useful information to ensure that you get the most out of the product.
-

1.1.1 Dangers

This Manual does not contain any information at the "Danger" level.

1.1.2 WARNING



-
- Device, cables and accessories must be inspected before use to guarantee their normal and safe operation.
 - This equipment can only be connected to the socket with ground protection. Please adopt a rechargeable battery instead of the socket as the power supply if the socket is not provided with a ground lead.
 - To prevent fire or explosion, do not operate the system in the presence of anesthetic, flammable or explosive materials.
 - Do not open the equipment casing as there is the impending danger of electric shock. Equipment maintenance and upgrades must be carried out by maintenance technicians whom are trained and licensed by the manufacturer. Moreover, the process must be done only after the AC power supply is disconnected. Maintenance carried out by individuals non-affiliated to the manufacturer or by non-licensed personnel may affect the safety, performance and function of the product.
 - When used with electrosurgery equipment, the safety of patients should be ensured.
 - The patient's clinical condition and the working condition of the syringe pump must be monitored carefully, and the alarm volume and alarm levels need to be set according to the actual needs. Operation and performance relying solely on the auditory alarm system alone is not sufficient, and setting the alarm at a low volume may endanger the patient. If the alarm volume is less than the surroundings volume, which can further lead to operators identify alarm mistakenly.
 - Please carefully install the power line and cables with various accessories to prevent the patient from choking or suffocation caused by entanglement of the cables or by electrical disturbance.
 - The packaging materials must be disposed of in compliance with local laws and regulations or the hospital policy on waste management. They must be kept out of the reach of children.
 - Infusion cannula knots, filter coagulation and occlusions arising from needle insertion can cause the pressure inside the syringe to rise during infusion. When this occurs, removing the occlusion can cause excessive
-

liquid to be infused into the patient, so appropriate measures should be taken.

- The pump should not be placed more than 100 cm above or below the level of the patient's heart. The smaller the height difference between the pump, the more accurate the pressure test in the infusion cannula will be.
 - This equipment has to be used with professional medical consumables, and its accuracy cannot be guaranteed when it is used with a syringe which is a non-standard consumable or a consumable without calibration, please contact the company for calibration service.
 - Do not touch the patient when connecting the peripheral equipment via the input/output signal ports to prevent patient leakage current from exceeding the requirements specified by the standard.
 - In the process of defibrillation, do not touch patient and other non-defibrillation equipments to prevent electric shock damage, and defibrillation will not affect the basic performance (such as infusion accuracy, alarm and signal transmission) of the pump.
-

1.1.3 CAUTION



- Use the accessories specified in this Operator's Manual to guarantee the patient's safety.
 - When equipment and the accessories exceed their recommended service life, they must be disposed of in accordance with local statutes or hospital regulations.
 - Electromagnetic fields may affect equipment performance. This makes it necessary for other equipment used in the vicinity of the pump to meet EMC standards. Mobile phones, X ray and MRI equipment are all potential interference sources because of their high-intensity electromagnetic radiation.
 - Before the equipment is connected to the power supply, check that the voltage and frequency of the power supply match the specifications on the label or in this Operator's Manual.
 - Please install and carry the equipment correctly to protect the equipment from damage from drops, impacts, violent shaking or other external mechanical forces.
 - Disposable accessories must be disposed of after use in accordance
-

with the relevant hospital regulations.

- Avoid direct sunshine, high temperatures and dampness.
 - Check the built-in battery before use to make sure it has sufficient power. Recharge the battery if necessary.
 - The syringe with the luer taper is recommended for use, which can effectively prevent patients from under current caused by the occurrence of the cannula to slip out when under tension.
-

1.1.4 NOTE

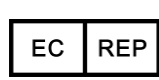
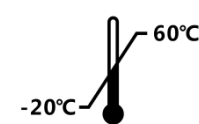
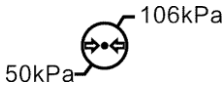
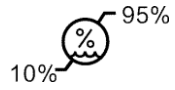
NOTE

- Install the equipment in a position where it can be easily accessed for inspection, operation and maintenance.
 - Keep this Operator's Manual near to the equipment for future ease of reference.
 - The software of the equipment is developed according to the software development demands of IEC60601-1 standard, which can minimize the possibility of the risk caused by program error.
 - This Operator's Manual describes the most complete functional configuration of the system. The product you are using may not have some of the settings or functions described herein.
 - Do not insert devices that are not specified by the manufacturer into the multifunction interfaces.
 - During infusion, the syringe pump can accurately control the rate, infusion volume and infusion time, and monitor the operation in real-time, to effectively prevent over currents, under currents and instances of backflow.
 - The device is not in touch with the drugs or patients directly. Thus, there is no need to process Biocompatibility test on it.
-

1.2 Equipment Symbols

The equipment you purchased may not provide you with all the following symbols.

	Refer to instruction manual/booklet		ON/OFF
	Alternating current		Battery
	Alarms		AUDIO PAUSED
	Clear/Back		Start; start of action
	Bolus		Confirm
	Stop		Menu
	Move up/Increase		Move down/Decrease
	Move left		Move right
	Syringe		Selected drug
	Night mode		Equipotentiality
	Wireless networking, normal communication with BeneFusion CS5 Infusion Supervision System		Wireless networking, no communication with BeneFusion CS5 Infusion Supervision System
	Wireless network is establishing a network connection		Current pump communicated successfully with BeneFusion DS5 Infusion Supervision System
	Current pump has downward relayed No.X pump		Current pump has forward relayed No.X pump
	Executed prescription		Pause prescription
	Executing prescription		Non-executed prescription

IP23	Protected against solid foreign objects with a diameter no less than 12.5mm and protected against spraying liquid water		DEFIBRILLATION-PROOF TYPE CF APPLIED PART
	Date of manufacture		Manufacturer
	THIS WAY UP		Keep dry
	Fragile, handle with care		STACKING LIMIT BY NUMBER
	Comply with the requirements of Directive 2012/19/EU Waste Electrical & Electronic Equipment		Non-ionizing electromagnetic radiation
	Input/output		Serial number
	Recovery/recyclable		Caution
	Environmentally-friendly use periods of electronic products (20 years)		Authorized representative in the European Community
	CE mark, comply with the requirements of the Council Directive 93/42/EEC (Medical Device Directive).		Package shall be kept between -20–60°C during transport
	Package shall be kept between 50–106 kPa during transport		Package shall be kept between 10%–95% humidity during transport

2 Theory of Operation

2.1 Overview

The Syringe Pump is used in conjunction with the syringe to control the dose of liquid infused into the patient's body.

The Syringe Pump is suitable for adults, children and newborns in clinical departments.

This Syringe Pump is expected to be used in institutes or units with healthcare capabilities. This includes but is not limited to: outpatient departments, emergency departments, wards, ICU, operating rooms, observation rooms, clinics, and nursing hospital.

The Syringe Pump primarily consists of a housing, a motor drive system, an input system, a memory system, a control system, a display system, a sensor-based monitor system and an alarm system.

Since some parts and functions are optional, the Syringe Pump you purchased may not contain these additional parts and their relevant functions.

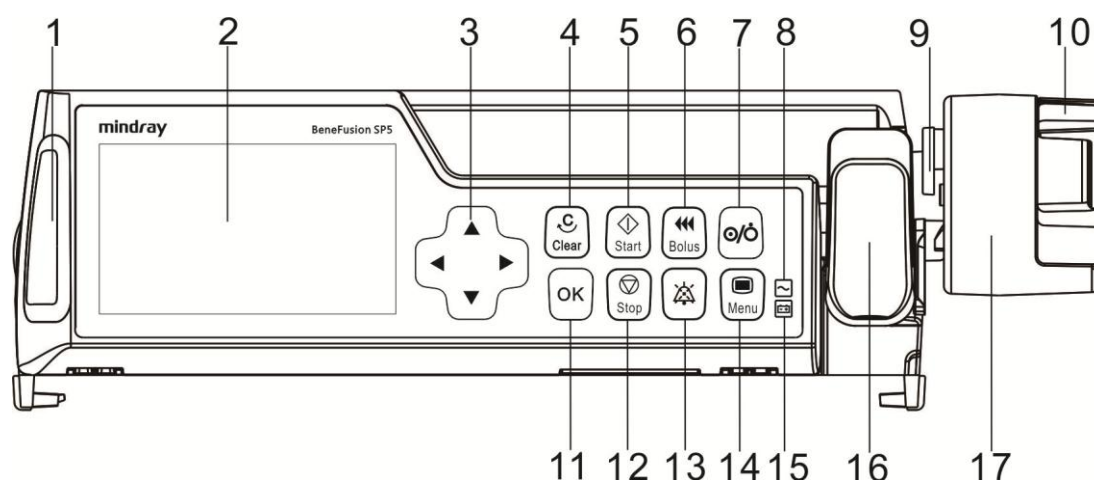
This syringe pump has the following functions and features:

- The syringe pump can accurately control the rate, infusion volume and infusion time, and monitor the operation in real-time, to effectively prevent over currents, under currents and instances of backflow.
- Good flow linearity.
- Changing the rate during operation.
- The syringe size can be identified automatically.
- Operation record, infusion record, daily alarm data export.
- Drug library, the current drug will be shown in operation screen
- Data saved, the memory can store up to 2000 records and 2,000 kinds of drugs.
- Manual bolus and auto bolus function.
- Standby function, keep the parameter set of previous standby.
- Purge function, air bubbles should be eliminated before the infusion.
- Dynamic pressure scanning function, monitor real-time pressure changes, as per pressure icon, to display the normal pressure status, near occlusion pressure status and occlusion alarm pressure status.
- Automatic locking and manual locking function, avoid the accident operation and misuse.
- Memory function, can record the last infusion parameter.

- Anti-bolus, the cannula pressure is released after occlusion.
- Power-on self-test function.
- Protected against defibrillation function.
- KVO function, to keep vein open.
- Fixing clamp can be installed horizontally and vertically, support at least 4 pumps overlapped.
- High-flux LED display.
- Alarm: visible alarm, audible alarm and alarm information, alarm volume with 10 levels selectable.
- Network communication: be connected with the central transfusion monitoring management system, infusion supervision system, and BB machine, through the approach of wireless networking. Data exported or upgraded by connecting to PC.
- Optional functions of the software comprise Rate Mode, Time Mode, Bodyweight Mode, TIVA Mode, Ramp Up/Down Mode, Sequential Mode, Loading Dose Mode, Micro-infusion Mode.

2.2 Host

2.2.1 Front View



1. Alarm light
The alarm light indicates different alarm levels in different colors and flash frequencies.
2. Display
Used for displaying infusion parameters and relevant content.
3. **<DIRECTION>**
Used for adjusting value, change lines and pages.

4. **<CLEAR/BACK>**

- Under non-setting status, indicate to returns to previous menu or operation.
- Under the setting status, indicate to clear the current set or cancel the edit.

5. **<START>**

After installing the syringe correctly and completing setting infusion parameters, press this key to start the infusion.

6. **<BOLUS >**

- During infusion, press this key to enter the Bolus Settings screen.
- When the pump is stopped, press this key to enter the purge settings screen.

7. **<POWER>**

- Used for turning power on, entering in standby state and turning off operations.
- When power off, press and hold (>3 s) the key.

8. AC indicator light

- On: The pump is connected to an AC power supply.
- Off: The pump is not connected to an AC power supply.

9. Clip

Clamps the syringe plunger.

10. Handle

Controls clip opening/closing and slider action.

11. **<OK>**

Used for confirming input operation and saving value.

12. **<STOP>**

During infusion, press this key in any non-setting interface to stop infusion. Infusion stops caused by alarms, such as occlusion and so on, press this key to cancel the alarm.

13. **<AUDIO PAUSED>**

Pauses alarm sound.

14. **<MENU>**

- Under non-operation status, used for switching [Main Menu] interface and other interfaces.
- Under operation status, press and hold this key to lock;in locked state, press and hold to unlock.

15. Battery indicator

- Steady green indicates that the battery is charging.
- Flashing indicates that the battery is providing power.
- Light off indicates that there is no battery or the equipment is turned off and not connected to an AC power supply.

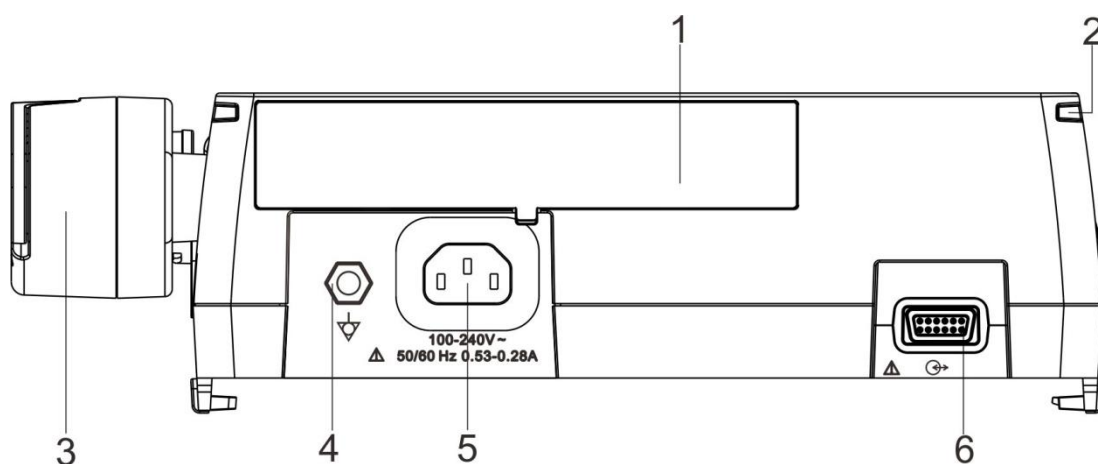
16. Spindle clamp

Used along with flange plate to secure syringe flange and clamps the syringe.

17. Slider

Secures syringe and drives plunger assembly.

2.2.2 Rear View



1. Battery box

2. Multi-channel pumps connect rail

Used for combining multi-channel pumps.

3. Slider

4. Equipotential port

When other device and syringe pump are used together, please adopt wires to connect other devices and equipotential port of the pump, to eliminate ground potential difference between different devices and ensure safety.

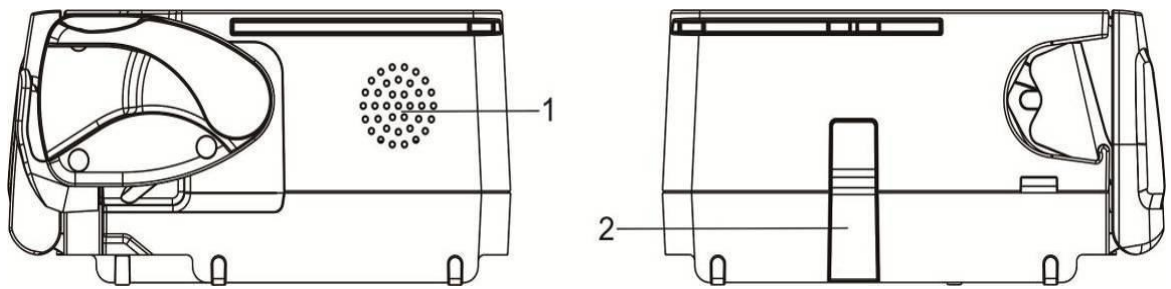
5. Alternating current power supply (AC) port

Connected by three-core-type power cord and AC power source.

6. Multifunction interface, which combines the following interface functions:

- DC power input interface
- RS232 interface
- Nurse call interface

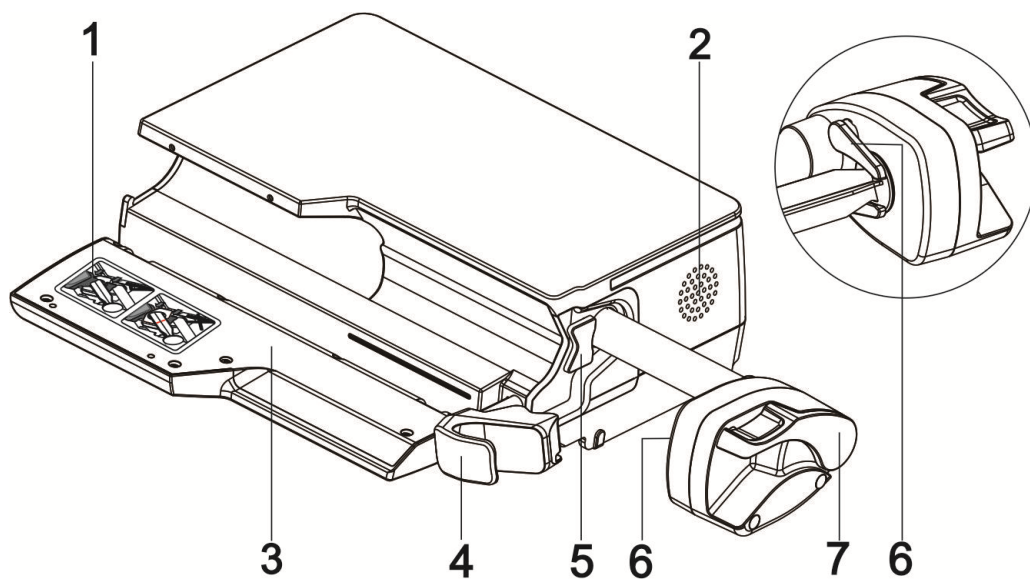
2.2.3 Side View



- 1. Speaker hole
- 2. Multi-channel pumps hasp

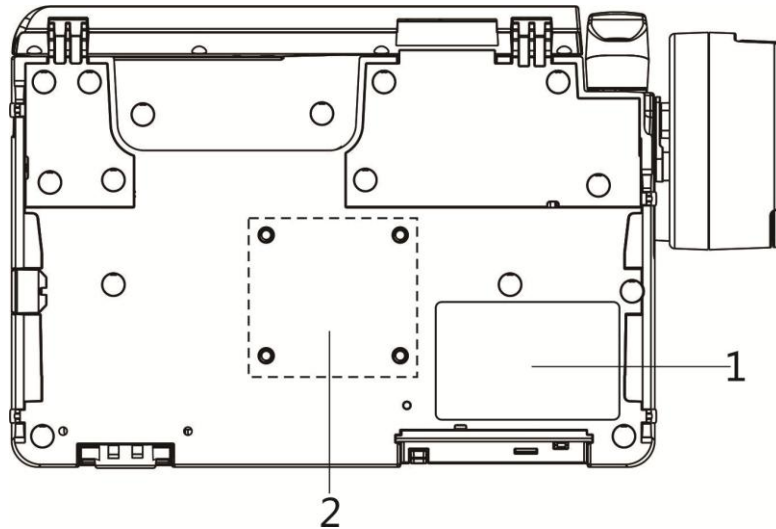
The multi-channel pumps are connected to secure the equipment.

2.2.4 Side View with the Door Opened



- 1. Warning label
- 2. Speaker hole
- 3. Door
- 4. Spindle clamp
- 5. Flange plate
- 6. Clip
- 7. Handle

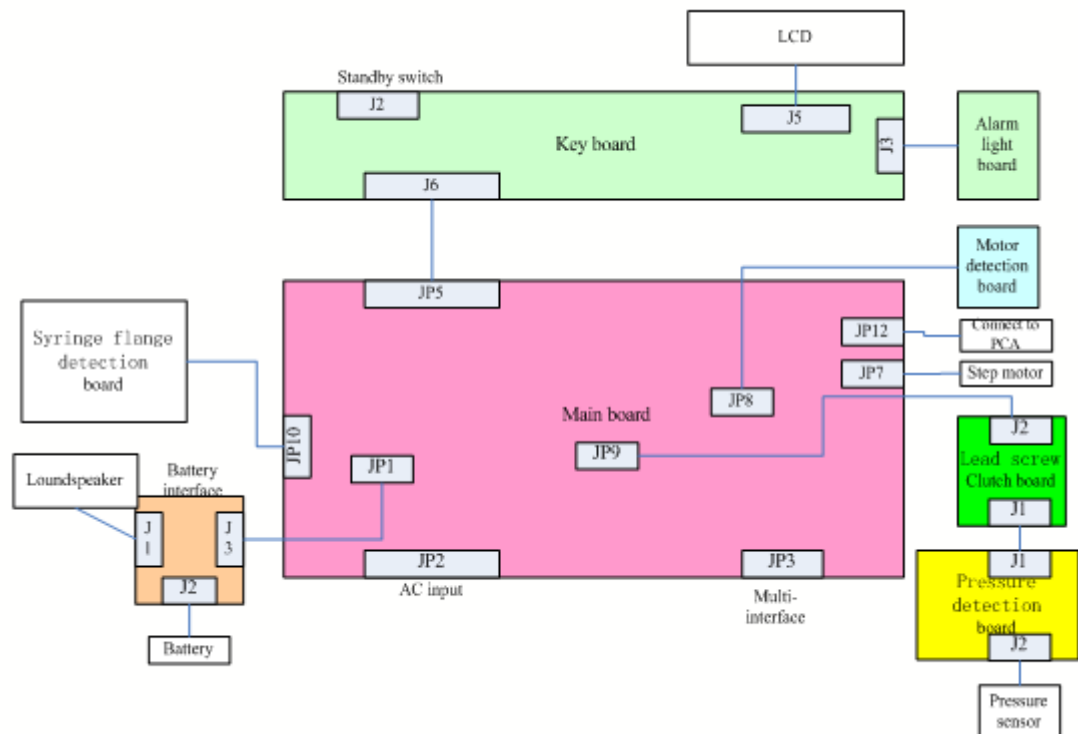
2.2.5 Bottom View



1. Product label
2. Fixed mounting holes

2.3 Hardware principle

SP5 hardware schematic diagram as shown below: including main board, keypad board, pressure board, etc.



2.3.1 Main board

SP5 main board includes 3 parts: power, master control and monitor.

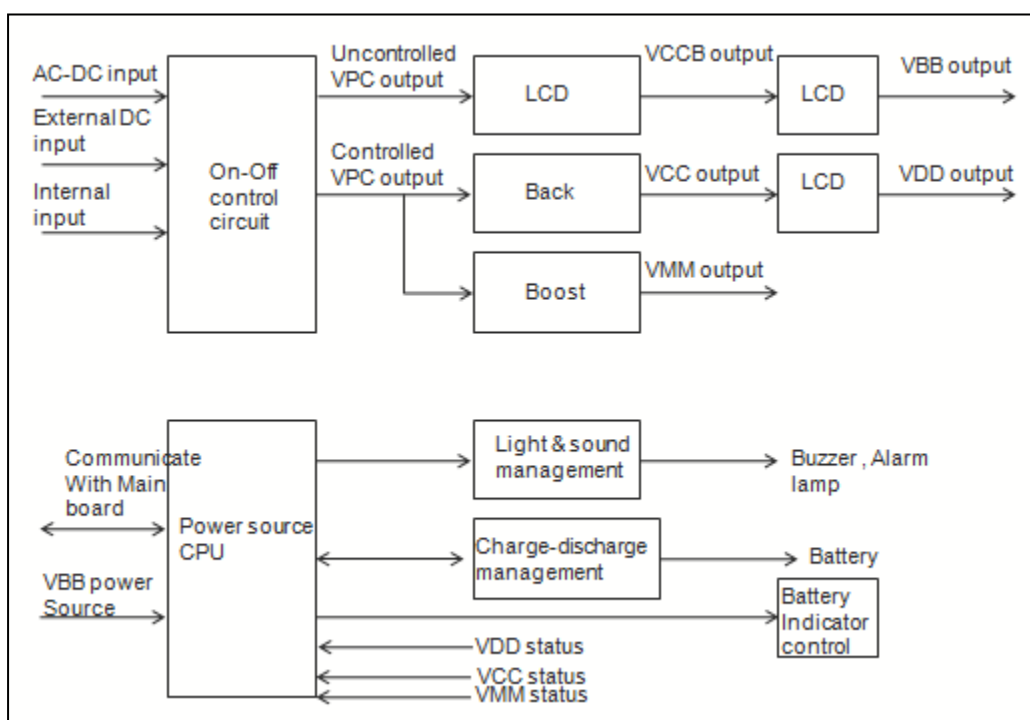
Power part includes ACDC conversion module, Switch startup & shutdown control circuit, DC power conversion circuit, power management CPU, standby sound-light alarm circuit, charge and discharge control circuit, etc.

The AC 100-240VAC, 50/60Hz input can be converted to DC 15V output by ACDC conversion module;

Switch control circuit can combine three-way merge power input into VPC, including the ACDC input, external DC input, internal battery; the ACDC input, external DC input priority is higher than the internal battery; VPC can be converted into VPP output by switch signal.

VPC is not controlled by switch signal, VPP is controlled by switch signal. VPC is reduced to VCCB (5V) by LDO circuit(U7 ZSR500GTA). VCCB is reduced to VBB(3.3V) by LDO circuit (U3 SPX5205); VPP is reduced to VCC(5V) by Buck circuit (U12 TPS5410D); VCC is reduced to VDD(3.3V) by LDO circuit (U13 SPX1117M3); VPP is increased to VMM(+24V) by Boost circuit (U26 LT3757EMSE).

The power management CPU is responsible for completing the switch control, under a single fault condition motor power-down protection, battery charging status monitoring, and the brightest state detection module power, the battery light control, and with the main control module CPU via the serial port. CPU Power Management selection Cortex M0 core chip LPC1112 (U10), powered by VBB supply.



Key signal testing points:

D5 positive-to-ground voltage(ACDC_IN) is tested about 15V

D6 positive-to-ground voltage(EDC_IN) is tested about 10V-16V

The voltage across of C154 (BAT_IN) is tested about 6.8V-8.4V V by multimeter

TP65 (VPC) is tested about 6.4V-15.6V by multimeter

TP66 (VPP) is tested about 6.4V-15.6V by multimeter

TP60 (VCCB) is tested about 4.5V-5.2V by multimeter

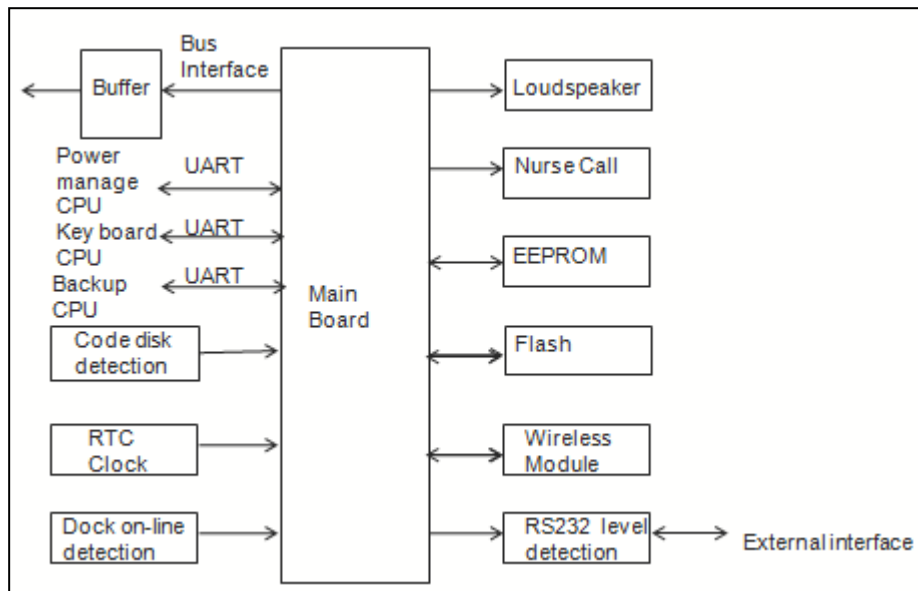
TP61 (VBB) is tested about 3.3V by multimeter

TP62 (VCC) is tested about 5.0 V by multimeter

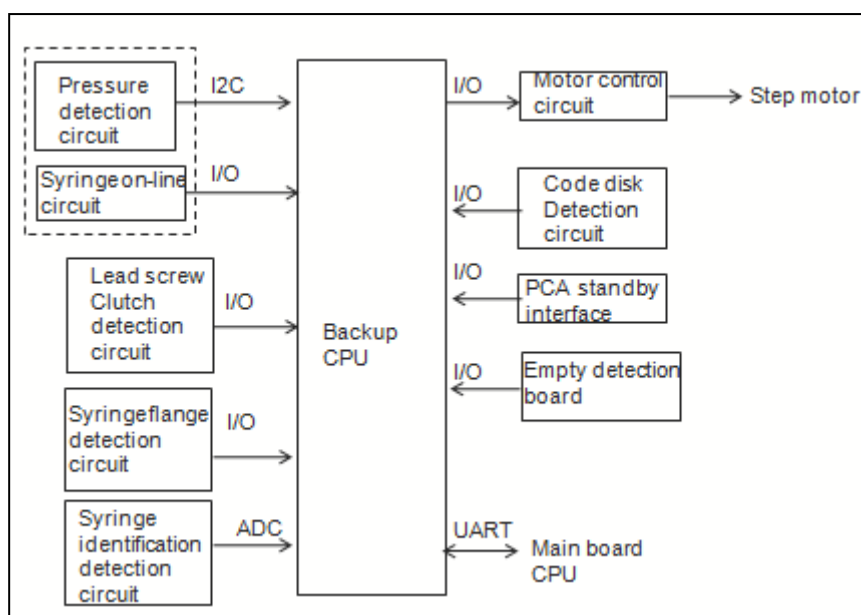
TP1 (VDD) is tested about 3.3V by multimeter

TP16 (VMM) is tested about 24V by multimeter

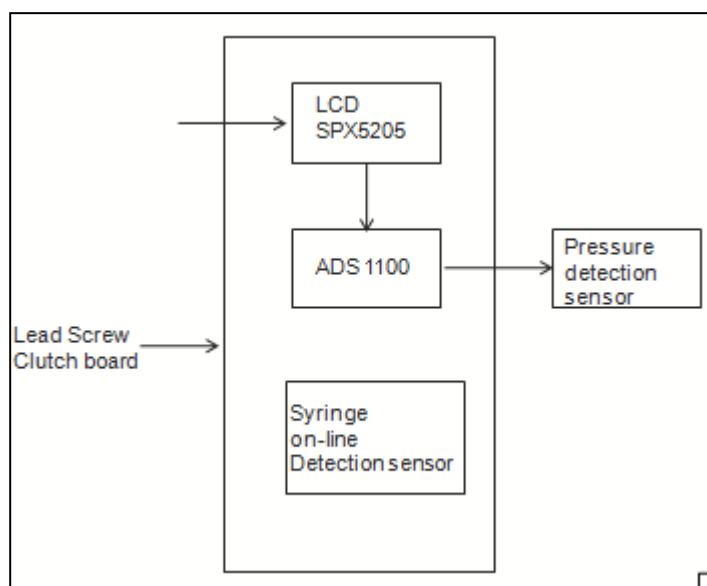
The main section is responsible for LCD display driver, the speaker alarm circuit, nurse call, Dock reign detection and power CPU, the key CPU, vice CPU UART communication, etc. The host CPU Cortex M3 core chip STM32F103VE (U14), using Intel 8080 system bus interface as the display driver interface.



Monitoring module is consist of vice CPU, stepper motor drive circuit, encoder detection circuit, the pressure detection circuit, the detection circuit reign syringe flange detection circuit. Vice CPU using STM32F100C8 (U1), stepper motor driver circuit using motor control chip TMC260-PA (U2). Dual encoder detection circuit using optocoupler, through the comparison of two sequences of pulses determines the motion direction of the motor by measuring the frequency of a pulse sequence determines the motor speed.

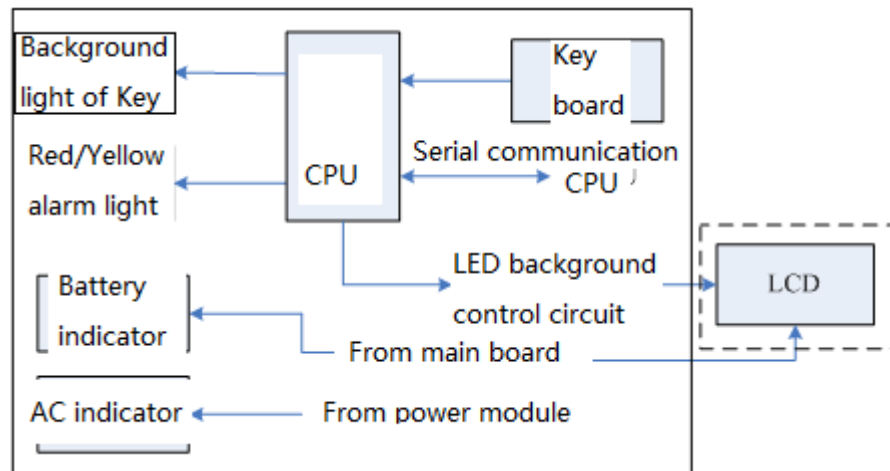


2.3.2 Pressure board



Pressure monitoring board includes a pressure sensing circuit and syringes reign detection circuit, the pressure detection circuit composed by the ADS1100 and SPX5205, SPX5205 will convert to +3.3 V +5 V output, +3.3 V power supply to the ADS1100 and pressure sensor; ADS1100 analog pressure sensor will input signal is converted into a digital signal, transmitted to the sub CPU through the I2C interface.

2.3.3 Keypad board



Key management using Cortex M0 core CPU chip LPC1112 (U4) is responsible for the keys (of 12 keys, including the power switch key directly lead to the power supply module CPU; other keys by the key management of the CPU processing) scan, alarm lamp drivers, LCD backlight control (8 levels adjustable) and other functions.

Key signal testing points:

TP10VDD 3.3V

TP11VCC 5.0V

TP12VPP

TP2LCD backlight power supply

TP20battery indicator light signal

TP21AC indicator light signal

3 Testing and Maintenance

3.1 Description

To ensure long reliable run of the syringe pump, the service people shall carry out periodical inspection, maintenance and testing. This section provides the basic testing methods for syringe pump. In this section, we have also recommended the suitable testing intervals and tools. The service people are required to repair and test the syringe pump as needed, and select the suitable testing tools.

The testing and inspection methods provided in this chapter are mainly used for verifying whether the syringe pump reaches the performance requirements. If the testing result does not meet the requirements, it indicates that some function module of the syringe pump is failed. In this case, immediate repair or replacement is required. If you have any other questions, please contact our After-sale Service Division.



CAUTION

- All the tests must be done by qualified service people.
- Before testing, the service people shall ensure whether the testing tools and connection wires are suitable. The service people shall have good skills on use of these testing tools.

3.1.1 Test Report

After finishing the test, the service people approved by Mindray Scientific are required to complete the following test report and send it back to the After-sale Service Division.

Test Equipment			
Name		Model / Number	Effective Date
Test Records			
NO.	Test Item	Test Point	Test Result
1			
2			
3			
4			
5			
Judgment			
If OK:		Tester:	Test Date:

3.1.2 Recommended Interval

The following tasks must be conducted by professional maintenance personnel approved by Mindray Scientific. Must clean and disinfect the device before the test or maintenance.

Check / Maintenance Item		Interval
Appearance Inspection		After the first installation or each reinstallation.
Power-On Self-Test (POST)		1. After the first installation or each reinstallation. 2. After each repair or replacement of the pump components.
Calibration		Every two years at least, using a new syringe brand. or when the user doubts that the syringe accuracy is incorrect.
Operation test		Every one year at least, or when the user doubts that the syringe flow alarm and pressure are incorrect.
Battery Test		1. Every year 2. Before repair of syringe pump or when the user doubts that the battery is the cause of trouble.
Electrical Safety Test	Electric Leakage Test of Enclosure	1. After repair or replacement of the power module 2. Every 2 years at least
	Electric Leakage Test of Patient	
Cleaning and Sterilizing		1. Every month recommended. 2. Thorough cleaning of the pump surface is required before or after a long storage.

3.2 Appearance Inspection

- 1) Pump casing clean, no scratching, installation tightly, sharking machine without internal residues.
- 2) Keypad operation smoothly.
- 3) Full label, correct information.
- 4) Standard configuration completely, power socket fixed tightly.
- 5) Make the vibration test on machine before operation.

3.3 Power-On Self-Test (POST)

- 1) After completing the syringe pump set up, it can enter system self-test, normal display, without any wrong information.
- 2) The system will give out a sound “di” and buzzer “di di”.
- 3) During power on, the alarm light will changed from red to yellow orderly.
- 4) Enter the memory mode parameter setting interface normally.

NOTE

-
- Six POST codes corresponding to the self-test items are: 1: reserved, unused temporarily; 2: Internal flash, external flash; 3: Communications detection (drive, power supply, keypad); 4: Alarm lights, buzzer; 5: Pressure coefficient (driver eepROM), 24V power supply; 6: reserved, unused temporarily.
 - system self-test failed, press the C key to enter the system
-

3.4 Calibration

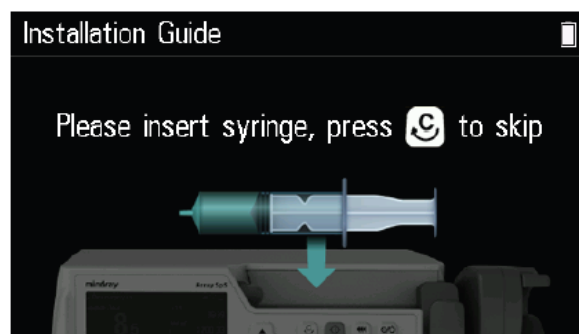
3.4.1 Syringe calibration

The purpose of Syringe calibration:

In order to improve the injection accuracy, make the calibration for the non-defined syringe brand, calibrated parameters including syringe diameter, full--scale length and empty position, the data of brand definition is empty in the factory; if the defined syringe brand is not calibrated, during using, the syringe brand list won't show this defined brand.

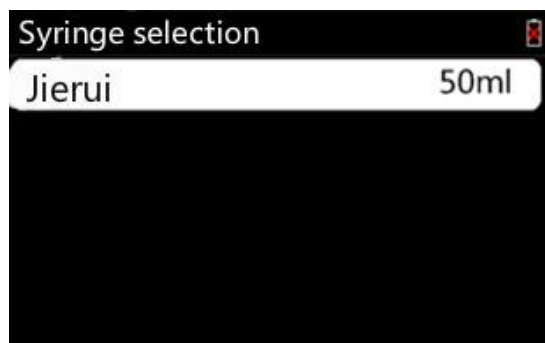
The steps for calibration are as follows:

1. No syringe installed:
 - a) After starting, it will enter the installation guide as shown below:



- b) Press “C” key to skip the step, enter the mode operation interface.
 - c) Press “menu” key to enter the main menu, choose “system maintenance”.
 - d) Input factory password “6686” to enter system maintenance menu.

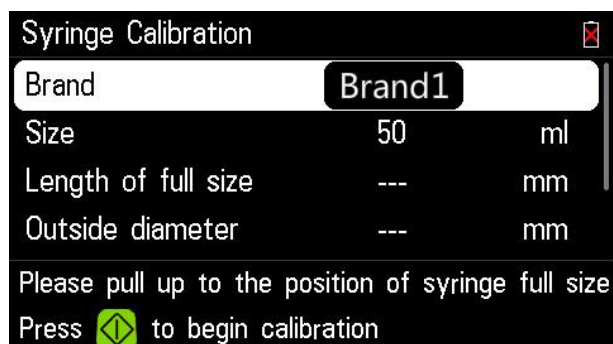
-
- e) Choose syringe calibration to enter.
 2. With a syringe installed:
 - a) After starting, enter brand select interface.



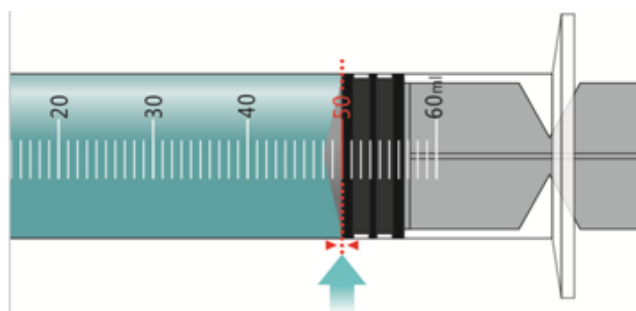
- b) Choose any brand, then enter memory mode operation menu.
- c) Press “menu” key to enter main menu, choose “system maintenance”.
- d) Input factory password “6686” to enter system maintenance menu.
- e) Choose syringe calibration to enter.

The calibration method of syringe:

- a) In system calibration interface, brand select “define 1” or “define 2”.
- b) Select the syringe specification, as shown below:



- c) Pull the syringe to the full dimensions and install them on the syringe pump, take 50ml syringe as an example, below refers to the full dimension of the 50 ml syringe:



- d) Press start key and the syringe pump starts calibration as shown below:

Syringe Calibration		
Brand	Brand1	
Size	50	ml
Length of full size	---	mm
Outside diameter	---	mm
Please pull up to the position of syringe full size		
Calibrating, press [stop icon] to stop calibration		

- e) The system will fill in the calibration data after the calibration is successful, and the screen prompts calibration completed.

Syringe Calibration		
Brand	Brand1	
Size	50	ml
Length of full size	80.94	mm
Outside diameter	30.8	mm
Please pull up to the position of syringe full size		
Calibration complete		

3.4.2 Pressure calibration

The purpose of pressure calibration:

It's to calibrate the pressure coefficient of non-built brand syringe, making occlusion pressure is more accurate, the users can also use without calibration, it won't have great effect on the accuracy of occlusion pressure, pressure calibration need to calibrate two pressure points, the default is 150mmHg and 950mmHg

Enter the pressure calibration:

1. Enter system maintenance, same as syringe calibration.
2. Under system maintenance menu, choose "pressure calibration".

Calibration method:

- a) Enter pressure calibration menu, choose "define 1" or "define 2" and size, as shown below:

Pressure Calibration		
Current AD...	0/0.00kgf/0.00mmHg	
Calibration rate	1000	r/h
Brand	Brand1	
Size	5	ml
Target cali.Value 1	150	mmHg
Press [start icon] to the first calibration		

- b) The syringe filled with fluid, connect one side of syringe to fluke, then install it to pump, press “start” key to start calibration.
- c) When fluke pressure value reach 150mmHg, press stop key, the first calibration completed.
- d) Continue to press start key to start the second calibration, when fluke pressure value reach 950mmHg, press stop key, the second calibration completed.
- e) Calibration completed, the system will fill in calibration data P value automatically, the reminder box will show “calibration completed”.

3.4.3 Displacement sensor calibration

The purpose of calibration:

Because there is mechanical error between short potentiometer (user Identity syringe size) and emptying position in each machine, in order to identify the outer diameter of the syringe and syringe emptying position more precise, each machine will calibrate the displacement sensor at the factory.

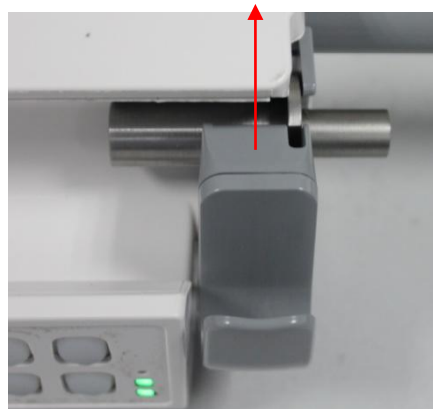
Calibration method:

- a) Under the system maintenance menu, enter the displacement sensor calibration, prompt install the standard component 1.

Sensor Calibration		
Current OD	1	mm
Cur.len.of empty...	2.5	mm
Standrad OD 1	13.5	mm
Standrad OD 2	32	mm
Stan.len.of emp...	21	mm
Please install the standrad clip 1,press  to...		

- b) Install standard component I correctly, press bolus key to calibrate the standard component 1.

Short potentiometer and empty tool: 13.5mm

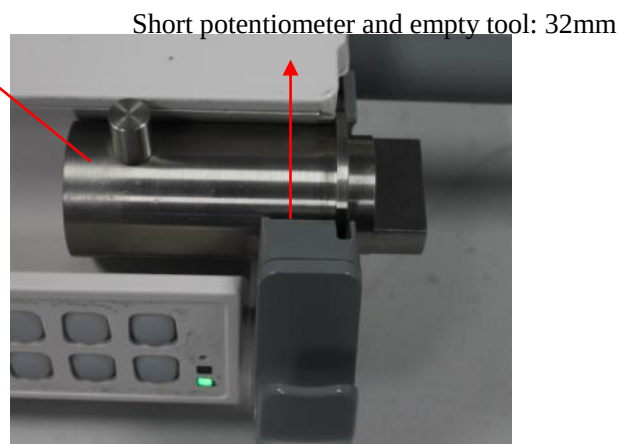


- c) After standard component 1 calibrated, the screen will prompt “please install standard component 2, press bolus key to calibrate”.

Sensor Calibration		
Current OD	1	mm
Cur.len.of empty...	2.5	mm
Standrad OD 1	13.5	mm
Standrad OD 2	32	mm
Stan.len.of emp...	21	mm
Please install the standrad clip2 press  to...		

- d) Install standard component 2 correctly, press bolus key to calibrate the standard component 2.

Registration
mast close to
the upper cover



- e) Calibration succeed, the system will prompt short potentiometer calibration successful.
- f) Calibration succeed, the current outer diameter will show that it's closed to the outer diameter of standard component 2.

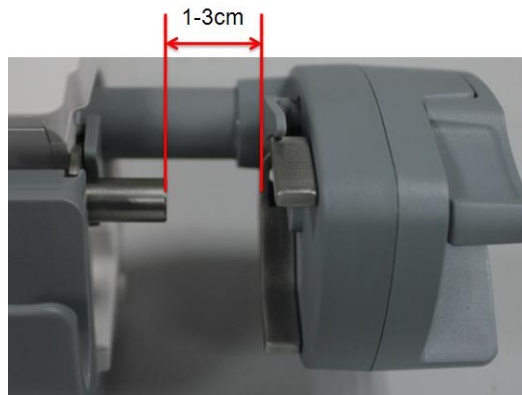
Sensor Calibration		
Current OD	32.1	mm
Cur.len.of empty...	2.5	mm
Standrad OD1	13.5	mm
Standard OD 2	32	mm
Stan.len.of emp...	21	mm
Short potentiometer calibration complete		

- g) Push the handle of slider, pull out the slider, and put the slider card into the jaws clamp around the limit between the clamps, as shown below.

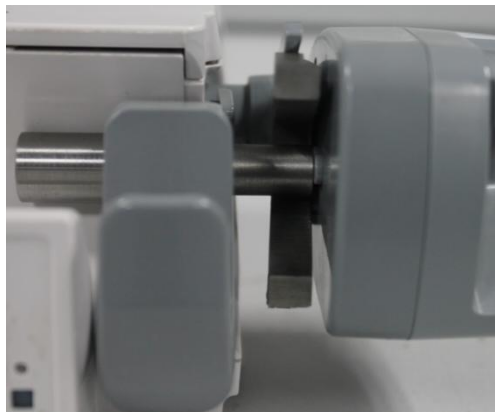
The limited clamp tool of slider card



- h) Move the cursor to the standard emptying position, install the standard 1 or standard 2, then pull out the slider, so that the top surface of the fixture and clamp jaws limit kept at a distance between 1-3cm, as shown below.



- i) Press “Start” key to start the calibration, the words “being calibrated, press the stop key to cancel the calibration” appears on the screen, after calibration completed, the words “emptying position calibration completed” appears on the screen when the calibration is completed, short potentiometer and emptying calibration fixture 13.5mm just withstood the knob of slider, as shown below.





- j) After calibration, the screen will show “calibration completed”.
- k) After emptying position calibration completed, the current location is displayed closed to emptying settings.

3.5 Operation test

General features:

- 1) For each key operation, MR pump should be able to press the “User's Guide” normal action or reaction, abnormal operation should not cause crashes.
- 2) Press the “User Manual” operation of the pump, check the sound and light alarm levels are normal, whether silence alarm is workable or not.
- 3) Press the “User Manual” operation of the pump and trigger the alarm, check whether the normal recording operation history and alarm.
- 4) After the injection parameters set correctly, press the start button to enter operation interface, run Fig standard display correctly.
- 5) Injection process runs Fig. Standard, the speed, the cumulative amount of pressure values displayed correctly

Networking function (only for the machine with network function):

- 1) Whether the central station and the pump power on, the networking should be successful.
- 2) The sending is normal between infusion data and central station.
- 3) The central station receive the pump status, infusion parameters, alarm information correctly.
- 4) If the connecting link is disconnected, the central station can indicate correctly, when restore the connection, the network will work properly.

3.6 Battery test

- 1) No battery inserted Uninstalled battery, low battery, the system will give a reminder.
- 2) When the battery capacity exhausts, the pump won't start or stop operation.

3.7 Electrical Safety Test

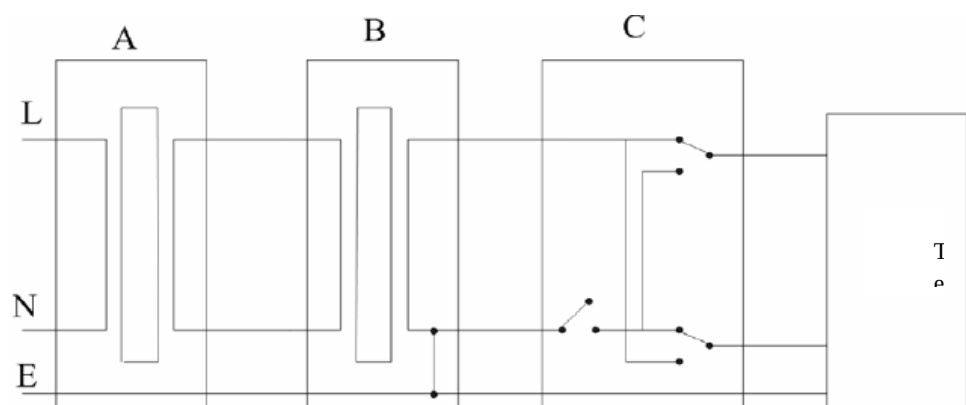
WARNING

- The electrical safety testing is intended for finding out the abnormal electrical hazards by using the accredited tester. Failure to discover these hazards might cause injury to the patients or operators.
 - The electrical safety testing may be done by using the commercially provided safety analyzer and other testing devices. The service people must guarantee the suitability, functional integrity and safety of these testing devices, while having good skills on their use.
 - The electrical safety testing shall follow the standards of the latest version, that is, EN 60601-1.
 - Please also observe the local laws on electrical safety testing, if any.
 - In the patient ward, all the equipments that are powered by city grid and connected to the medical instrument must comply with IEC 60601-1 standard. Meanwhile, electrical safety testing must be done to these equipments according to the testing interval of the tester.
-

The electrical safety testing is intended for finding the potential hazards that might cause electrical safety risk to the patients, operators and repairmen. Please carry out the electrical safety testing under normal environment (including: temperature, humidity and atmospheric pressure).

The electrical safety testing described in this chapter is based on 601 safety analyzer. Different safety analyzers might be used in different regions. Please ensure the suitability of your electrical safety testing program.

The equipment connection schematic is as below:



A: AC Power (programmable, frequency adjustable); B: Isolation transformer on the leakage current test tool; C: Safety test device

Testing tools:

- Safety analyzer
- Isolation transformer

3.7.1 Electric Leakage Test of Enclosure

1. Connect the 601 safety analyzer to 264 VAC/60 Hz power source.
2. Connect the applied part of the tested equipment by using the applied part connection tooling. The SUM end of the connection tooling for applied part is connected to the RA end of the safety analyzer.
3. Via power cord, connect the tested equipment to the auxiliary power output port on 601 safety analyzer.
4. Connect one end of the red testing wire to the “Red input terminal” on the safety analyzer, and clip another end tightly onto the metal foil on the surface of tested equipment enclosure.
5. Turn on the power source of 601 safety analyzer and press “5-Enclosureleakage” on the panel of 601 safety analyzer to enter the enclosure leakage current testing interface.
6. The enclosure leakage current shall not exceed 100 μ A under normal conditions and shall not exceed 300 μ A in event of a single fault.

3.7.2 Electric leakage test of earth

1. Connect the 601 safety analyzer to 264 VAC/60 Hz power source
2. Connect the applied part of the tested equipment is connected to the RA end of the safety analyzer.
3. Via power cord, connect the tested equipment to the auxiliary power output port on 601 safety analyzer.
4. Turn on the power source of 601 safety analyzer and press “4-Earth leakage” on the panel of 601 safety analyzer to enter the earth leakage current testing interface.
5. The earth leakage current shall not exceed 500 μ A under normal conditions and shall not exceed 1000 μ A in event of a single fault.

3.7.3 Electric Leakage Test of Patient

1. Connect the 601 safety analyzer to 264 VAC/60 Hz power source.
2. Connect the applied part of the tested equipment by using the applied part connection tooling. The SUM end of the connection tooling for the applied part is connected to the RA end of the safety analyzer.
3. Via power cord, connect the tested equipment to the auxiliary power output port on 601 safety analyzer.
4. Turn on the power source of 601 safety analyzer and press “6-Patient leakage” on the panel of 601 safety analyzer.
5. Keep pressing “APPLIED PART” key to select AC or DC measuring. If DC is selected, “DC” is displayed behind the limit value.
6. The patient leakage current shall not exceed 10 μ A under normal conditions and shall not exceed 50 μ A in case of a single fault.

If the electrical safety testing fails, please contact our technical support people.

3.8 Cleaning and Disinfection

3.8.1 Cleaning

The pump should be cleaned regularly. If operating in dirty or sandy areas, cleaning should be more frequent. Before cleaning, please consult or refer to the hospital's specific regulations concerning medical device cleaning.

The following are detergents available for selection:

- Hydrogen peroxide (3%)
- Ethanol (70%)
- Isopropanol (70%)

When cleaning the device:

1. Turn off the pump and disconnect the AC power source line.
2. Wipe the display screen after soft cotton balls absorb an appropriate amount of detergent.
3. Use a piece of soft cloth which absorbs a modest amount of cleaning agent to wipe the surface of the device.
4. When necessary, use a piece of cloth to wipe off any excess cleaning agents.
5. Place the equipment in a cool and ventilated environment to dry.

3.8.2 Disinfection

The operation of disinfection may cause certain damage to the syringe pump. You are recommended to disinfect only when it is necessary in your desired maintenance plan. Clean the equipment before disinfection.

The recommended disinfectants include: ethano 70%, isopropanol 70%, glutaraldehyde-type 2% liquid disinfectant.

CAUTION

- **Never use EtO or formaldehyde for disinfection.**
 - **Do not conduct high pressure or high temperature disinfection for the syringe pump and its accessories.**
-

4 Troubleshooting

4.1 Overview

This chapter provides categorized description of the syringe pump troubles by fault position and symptom. When eliminating the trouble, please refer to the corresponding fault sheet and check, locate and eliminate the fault in order.

The recommended methods in this chapter can help you solve most of the equipment troubles, but exclusive of any problem that might occur. In case of any fault not described in this chapter, please contact our After-sale Service Division.

4.2 Replacing Components

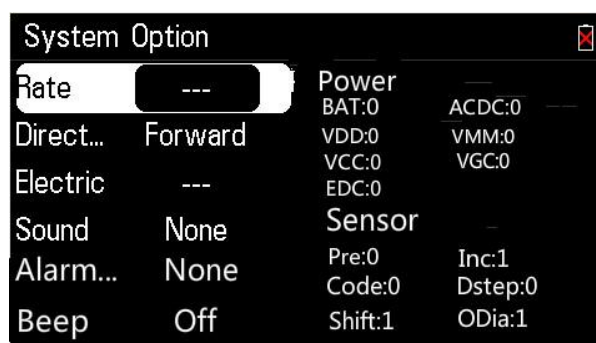
You may replace the circuit board set and other main components or parts of the syringe pump. Once you have located the circuit board set with trouble, you may replace it according to the steps described in **Chapter 5 Repair and Disassembly**. Then, check if the trouble has been eliminated, or if the syringe pump can pass the relevant testing. If the trouble is eliminated, it indicates that the original circuit board set is damaged. In this case, please return it to our company for repair. If the trouble remains, please reinstall the original circuit board set and try eliminating the trouble from other possible causes.

To know and obtain the components you need to replace, please refer to **Chapter 6 Parts**.

4.3 Check syringe pump status

The instruction of AD value access and checkout content

1. The instruction of AD value screen: System Maintenance-password-sensor AD value 2AD value.
2. The define of each AD value, as shown below:



- BAT: The battery value
- VDD: 5V voltage

- VCC: 5V voltage
- EDC: DC power supply, Multi-channel Buffered Serial Port
- ACDC: 15V regulated power supply
- VMM: Motor voltage 24V
- VGC: Gold capacitance voltage 5V
- Pre: Pressure Value
- Coder: Encoder light coupling
- Shift: Displacement of the coupling of light
- Inc: Syringe installation, wing handle, nut
- Dstep: Lose count, Encoder testing and the actual value difference
- ODia: Short potentiometer AD value

3. The value of ins shows the following 3 sensor AD value:

/	Natural state	AD value based on natural state	Sensor interface corresponds to which order of several binary
Convex edge detection optocoupler (pull the handle at short potentiometer)	Retain	1	Highest level
Screw clutch optocoupler (push the handle of slider)	Retain	0	Middle level
Pressure detection optocoupler (install syringe)	Retain	1	Lowest level

4. ODia means AD value of short potentiometer, pull the short potentiometer handle smoothly, the ODia value will be larger, the scope should be between 0-4095;
5. ShiftTable shown emptying detection optocoupler AD value (ie: similar long potentiometer SK pump), consisting of a total of four optocouplers, four optocoupler corresponding four binary digits, the machine leftmost lowest position, each optocoupler outputs AD when the block is 1, the output is not blocked when AD is 0;
6. PreTable AD showing pressure sensor output value, when the pressure to the pressure sensor, this value should be increased;
7. CodeTable code shown optocoupler output disc detection AD value, under normal circumstances, the motor rotates, this value should be between 0,1,2,3 cycle;

4.4 Fault list

SP5 in the transportation, storage, use, due to various factors (such as the grid voltage instability, changes in ambient temperature, drop and shock, component aging, etc.) influence may malfunction and affect the normal use. At this point there should be some component level repair of electronic medical equipment maintenance experienced professionals refer to the following fault classification. The so-called after the component level repair refers to the parts of the analysis, substitution, commissioning, and fault locate the parts of the instrument, such as the power supply section, Master control part, monitoring part, LCD and so on. Component repair itself that component level repair, must have extensive experience in the maintenance by the manufacturer's engineers use specialized equipment, carried out under specific circumstances and conditions.

4.4.1 Alarm information and solution

Please refer to the Benefusion SP5 Operator's Manual.

Machine malfunction

Symptom	Possible Cause	Fault Elimination
No AC power supply is connected, without any display after power on, the battery indicator light isn't on, the buzzer won't ring	1.No power in battery; 2.Bad connect with the wire of battery interface board; 3.Bad connect with keypad board FPC 4.Main board damaged	1.Replace the new battery 2.Repair or replace the wire of battery interface board; 3.Reinstall or replace the FPC wire; 4.Replace the main board
No AC power supply is connected, without any display after power on, the battery indicator light is on, the buzzer will ring	1.Display damaged; 2.Short circuiting damaged	1.Display troubleshooting; 2.Check the main board and other boards
AC power supply is connected, the AC indicator light isn't on	1.Bad connect with the wire of keypad board FPC; 2.Bad connect with the wire of AC power supply; 3.Malfunction of ACDC module.	1.Repair or replace the FPC wire; 2.Repair or replace the wire of AC power supply; 3.Repair or replace the ACDC module
Didn't pass the Self-inspection 3, alarm	1.Bad connect with keypad board FPC wire; 2.Malfunction of main board; 3.Malfunction of keypad board	1.Repair or replace the FPC wire; 2.Replace main board; 3.Replace keypad board
Didn't pass the Self-inspection 5, alarm	1.Bad connect with FPC wire of lead screw clutch board; 2.Bad connect with 6pin wire	1.Repair or replace the FPC wire of lead screw clutch board 2.Repair or replace the 6pin

Symptom	Possible Cause	Fault Elimination
	3.Malfunction of pressure sensor	wire 3.Repair or replace the pressure sensor
Cannot be recharged	1.Bad connect with wire of battery interface board; 2.Malfunction of battery; 3.Malfunction of main board	1.Repair or replace the wire of battery interface board; 2.Replace battery; 3.Replace main board
Some operation or measure function failed	Main board or specific parts damaged	Check the main board and specific parts

4.4.2 Screen malfunction

Symptom	Possible Cause	Fault Elimination
Power on normally, but with blank screen or during operation with white screen, or blank screen	1.Bad connect with keypad board FPC wire; 2.Malfunction of display 3.Malfunction of keypad board 4.Malfunction of main board	1.Repair or replace the FPC wire; 2.Replace the display 3.Replace keypad board; 4.Replace main board
The letter or pictures shown wrong	1.Resource file burning error 2.Malfunction of main board	1.Burning the resource file again 2.Replace main board

4.4.3 Operation, alarm fault

Symptom	Possible Cause	Fault Elimination
Keypad invalid	1.Bad connect with keypad board FPC wire; 2.Malfunction of keypad board	1.Repair or replace the FPC wire; 2.Replace keypad board
Buzzer sound invalid	1.Malfunction of main board	1.Replace main board
Speaker failure or hoarse voice	1.Malfunction of speaker; 2.Repair or replace the wire of battery interface board; 3.Malfunction of speaker alarm circuit	1.Repair the wire or replace speaker assembly; 2.Repair or replace the wire of battery interface board; 3.Check the damaged circuit
Red alarm indicator invalid or yellow alarm indicator invalid	1.Malfunction of alarm light board; 2.Bad connect with keypad board FPC wire; 3.Malfunction of keypad board 4.Malfunction of main board	1.Repair the wire or replace the alarm light board 2.Repair or replace the FPC wire; 3.Replace keypad board 4.Replace main board
Button backlight failure	1.Bad connect with keypad board FPC wire; 2.Malfunction of keypad board 3.Malfunction of main board	1.Repair or replace the FPC wire; 2.Replace keypad board; 3.Replace main board

4.4.4 Monitor fault

Symptom	Possible Cause	Fault Elimination
keep the same pressure value	1.Malfunction of pressure sensor 2.Malfunction of pressure test board	1.Replace pressure sensor 2.Replace pressure test board
Pressure over range	1.Malfunction of pressure sensor 2.Malfunction of pressure test board	1.Replace pressure sensor 2.Replace pressure test board
Can't identify the syringes	1.Short potentiometer slip handle and mechanical card sets out 2.Malfunction of short potentiometer	1.Mechanical card sets short potentiometer slip handle; 2.Replace short potentiometer
Malfunction of pump motor	1.Malfunction of motor test board; 2.Bad connect with motor wire 3.Malfunction of stepper motor circuit 4.Malfunction of pump motor;	1.Replace motor test board 2.Repair or replace the motor wire; 3.Replace main board; 4.Replace pump motor;
Detecting faulty in syringe placing.	1.Bad connect with screw clutch FPC; 2.Bad connect with 6Pin wire; 3.Malfunction of pressure test board	1.Repair or replace the FPC wire; 2.Repair or replace the 6Pin wire; 3.Replace pressure test board
Screw clutch fault detection	1.Bad connect with screw clutch FPC wire; 2.Malfunction of screw clutch board	1.Repair or replace the FPC wire; 2.Replace screw clutch board;
Flange detect failures	Malfunction of flange detect	Repair wire or replace flange detect board

5 Repair and Disassembly

NOTE

- Be sure to wear the antistatic wrist strap during the whole process of repair and disassembly.
-

5.1 Tool preparation

You might need the following tools during disassembly and replacement of the components:

- ◆ Crosshead screwdriver (Size PH1)
- ◆ Forceps
- ◆ Sharp nose pliers
- ◆ Side cutter
- ◆ Hex spanner 3mm
- ◆ Hex spanner 1.5mm
- ◆ Wrist strap

5.2 Preparations for Disassembly

Before disassembly of the syringe pump, please stop infusion to the patients and shut down the syringe pump, while disconnecting all the accessories and external equipments.

CAUTION

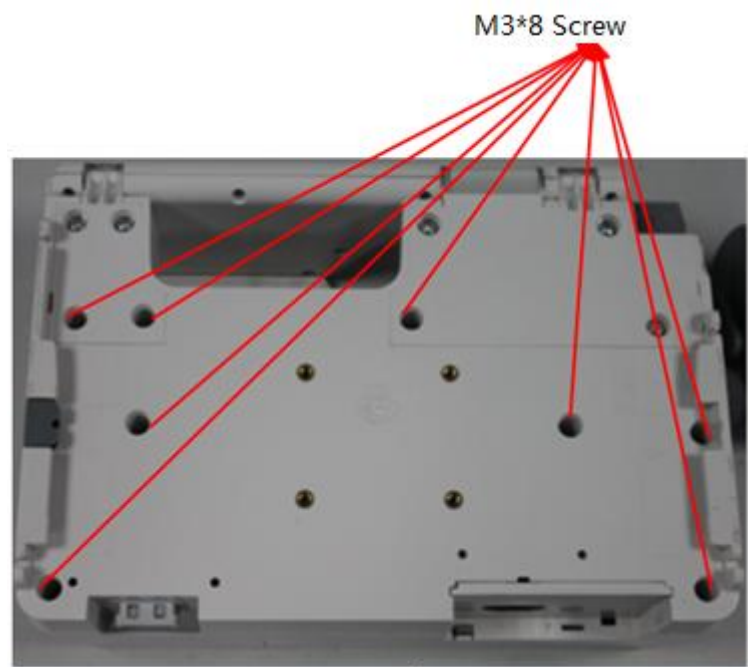
- Please eliminate the static electricity before disassembly. To avoid damage to the components, please wear personal protective equipment like wrist strap or electrostatic gloves when disassembling some components with electrostatic sensitive label.
 - During reinstallation, take care to connect the wires properly and place them in good order, so as to avoid the short circuiting due to damage of the connection wires.
 - Take care to select the appropriate screws during reinstallation. Forcible tightening of the incorrect screws might cause damage to the equipment. And during use after reinstallation, the screws or components might fall off, thus causing unpredictable damage or injury to the product or human body.
 - Disassemble the equipment in correct order. Incorrect order and forcible assembly might cause irreversible damage to the equipment.
-

-
- Before separating the components, please be sure that all connections have been plugged off. During separation, take care not to break the connection wire or damage the connector.
 - Please store the removed screws and other parts by category for easy reach during reinstallation. And avoid dropping, contaminating or losing them.
-

5.3 Disassembly Steps

5.3.1 Disassembly of UP cover assembly

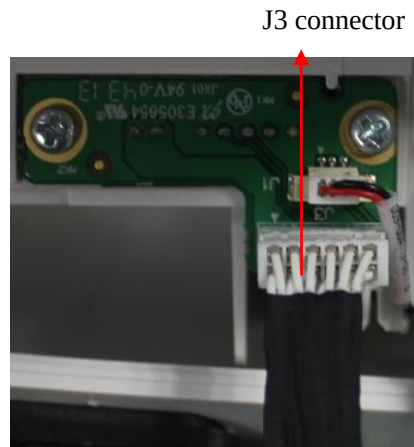
1. As shown below: remove 8 double gasket screws (M3x8) fixing the UP cover.



2. Vertically upward pull up cover assembly, as shown below:



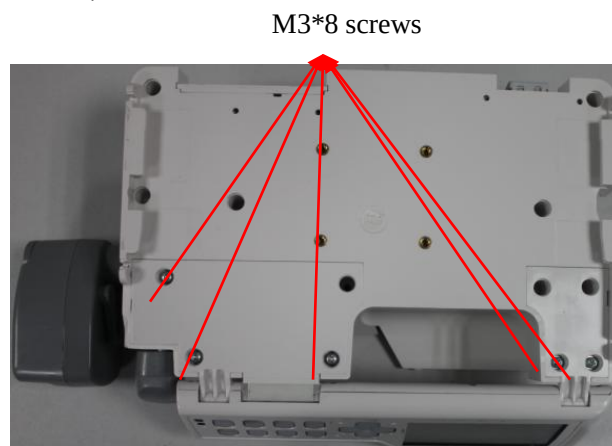
-
3. Remove the battery adapter board wire from the J3 connector on the battery adapter board PCBA, as shown below:



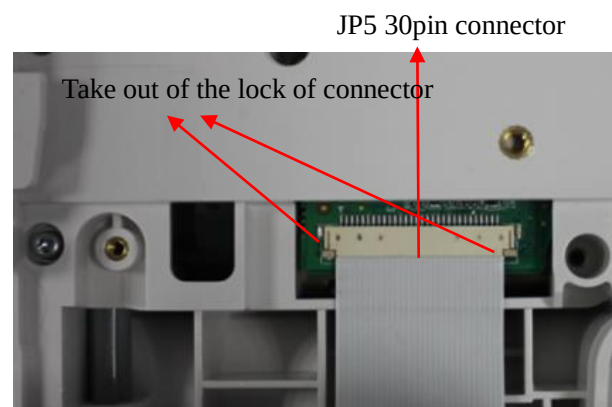
4. Then remove the up cover assembly

5.3.2 Door assembly

1. Remove 5 screws M3x8, as shown below:



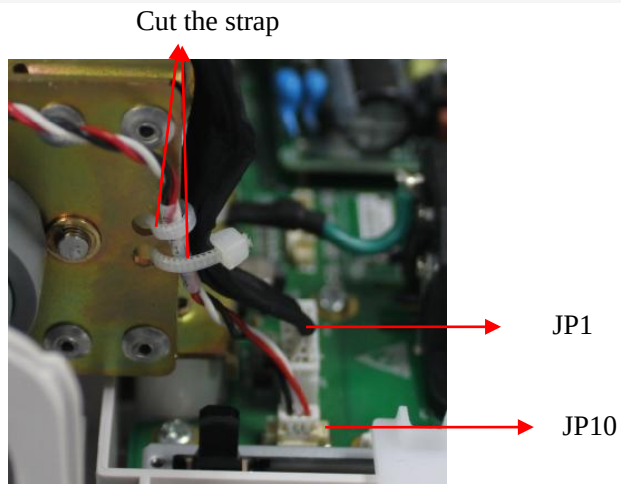
2. Take out of the lock of keypad board 30pin connector, separate the door assembly and main board, as shown below:



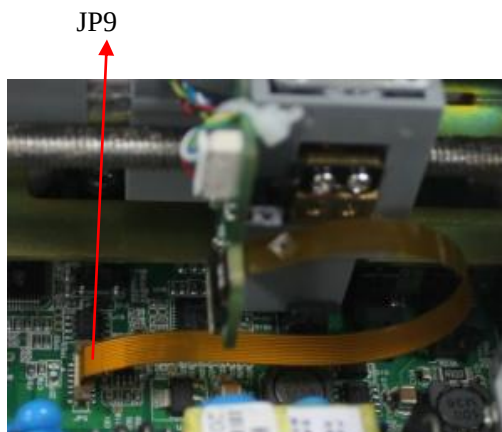
3. Remove the door assembly from host machine.

5.3.3 Disassembly of pump body assembly

1. Cut the strap, remove the convex detection wire and the wire of battery adapter board from JP1 and JP10 in main board, as shown below:



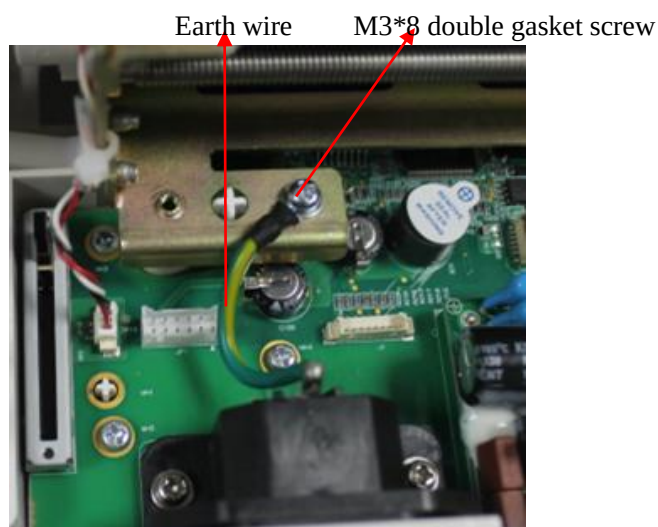
2. Take out of the screw clutch test FPC wire from the JP9 in main board, as shown below:



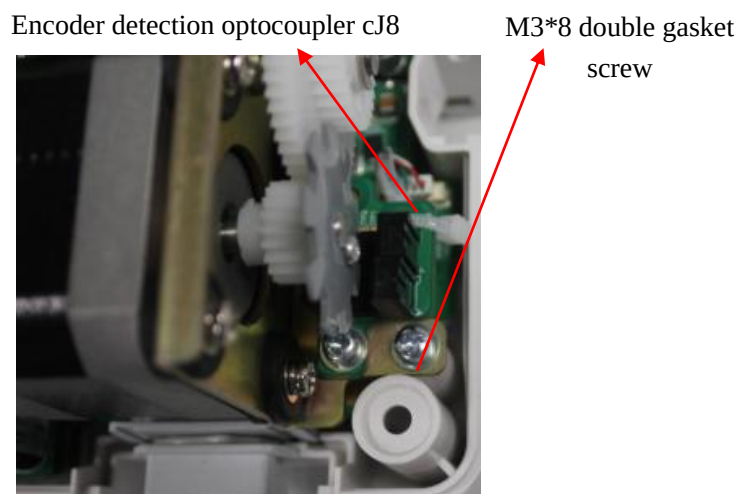
3. Remove 1 screw M3x8, as shown below:



-
4. Remove a double gasket screw M3x8, as shown below:



5. Remove the encoder detection optocoupler cable and motor cable from the J8, J7 connector on the main board, take out of the double gasket screw M3x8, as shown below:



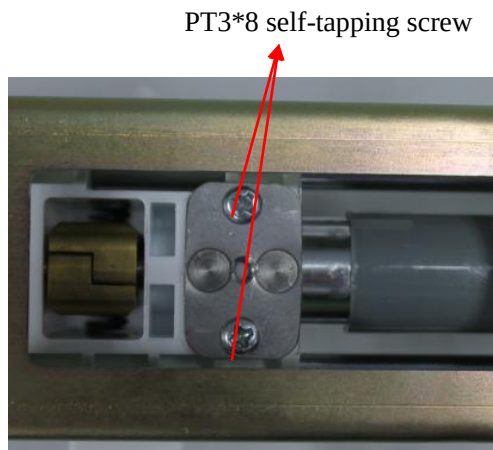
6. Remove the pump body assembly from rear cover.

5.3.4 Disassembly of rear cover assembly

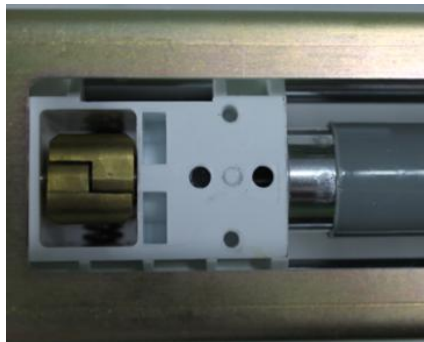
Remove up cover assembly, door assembly and pump body assembly, the rest is rear cover assembly.

5.3.5 Removal of push box assembly, convex detection assembly, pump body transmission assembly

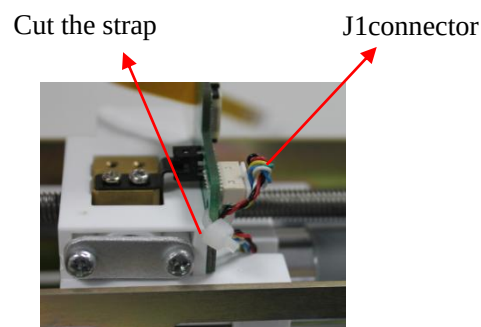
1. Remove 2 PT3x8 self-tapping screw, as shown below:



2. Take out of the hollow shaft fixing piece, as shown below:



3. Cut the strap in nut clutch detection board PCBA by Side cutter, then remove pressure detection wire from J1 connector of nut clutch detection board PCBA, as shown below:



4. Remove the slider assembly from sliding block assembly in force, then get the push box assembly, convex detection assembly, pump body transmission assembly

5.3.6 Disassembly of pressure sensor assembly

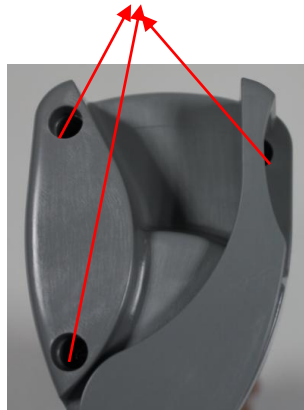
Note: Remove the pressure sensor assembly, make it directly in whole machine, and don't need to take out of the slider assembly.

1. Point the 3 slider rubber plugs by forceps, as shown below:



2. Loose 3 PT3×8 self-tapping screw, as shown below:

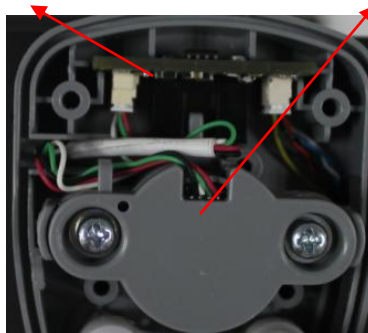
PT3*8 self-tapping screw



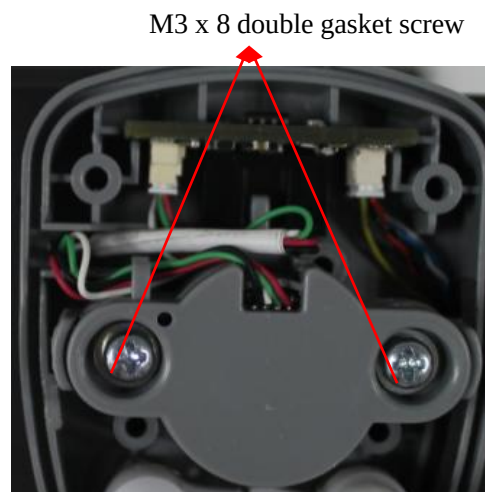
3. Remove the back cover of slider, then you can see the pressure sensor and pressure detection board PCBA, as shown below:

Pressure detection board PCBA

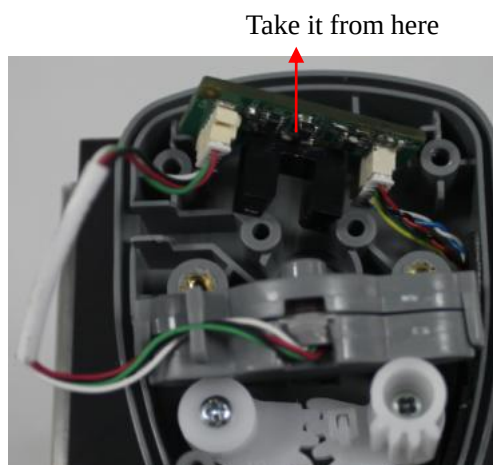
Pressure sensor



-
4. Take out of 2 M3x8 double gasket screw in pressure sensor assembly, as shown below:



5. Take out of the pressure detection board PCBA from slider, as shown below:



6. Pressure sensor assembly, as shown below:



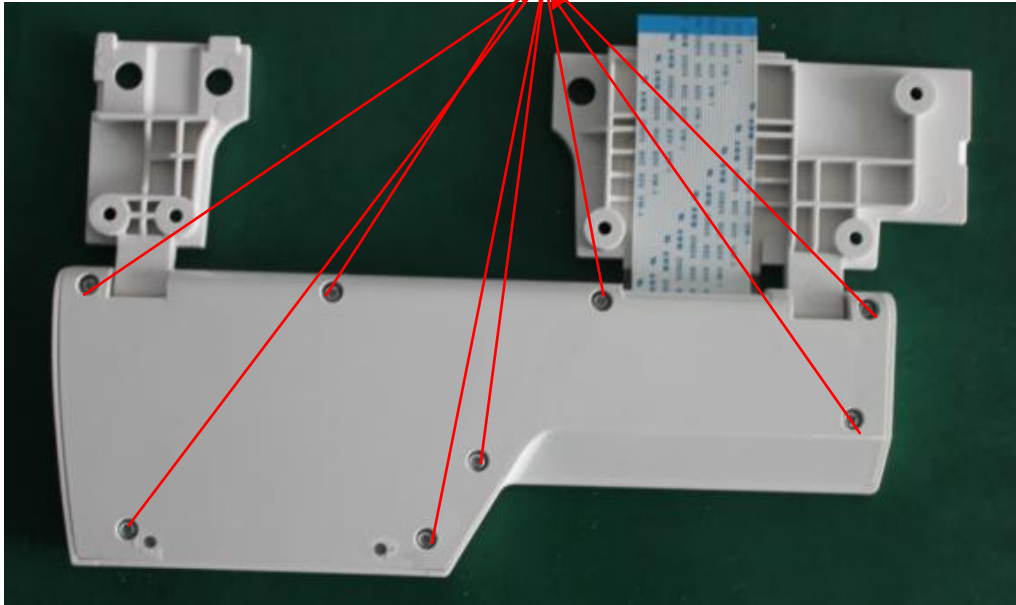
Notes: As show above, the assembly should be as a Min maintenance unit, if there is one material damaged, need to replace one assembly completely.

5.3.7 Disassembly of screen, keypad board PCBA

Note: Disassembly of screen, keypad board PCBA, make it directly in whole machine; don't need to take out of door assembly

1. Remove 8 PT2.6×6 self-tapping screw, as shown below:
- 2.

PT2.6×6 self-tapping screw



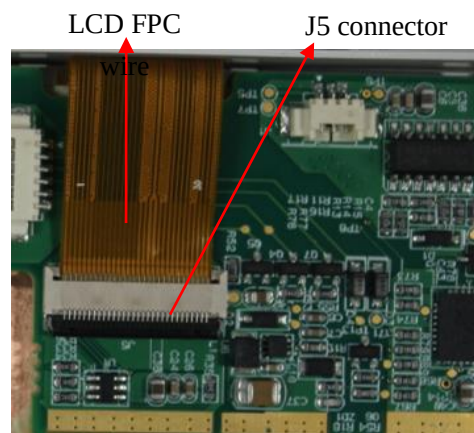
3. Remove the alarm indicator board PCBA from J3 connector of keypad board PCBA, as shown below:

Alarm indicator board PCBA

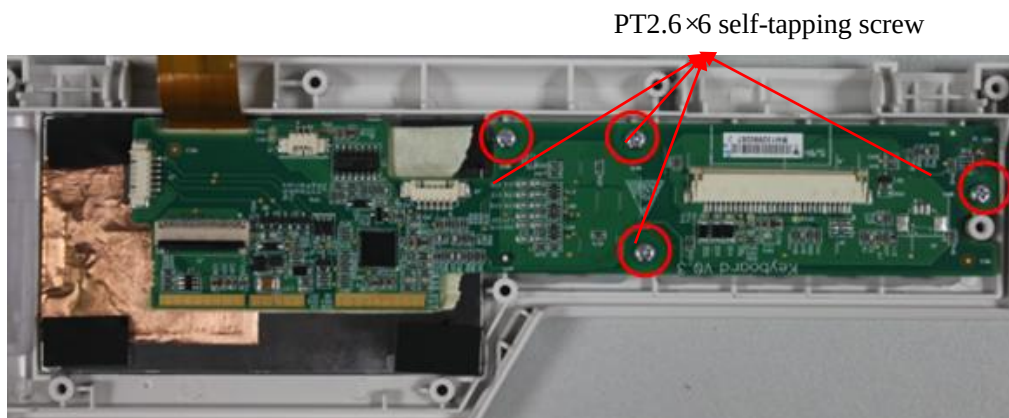
J3connector



-
4. Remove FPC wire of LCD from J5 connector of keypad board, as shown below:



5. Remove 4 PT2.6×6 self-tapping screw from keypad board PCBA, as shown below:

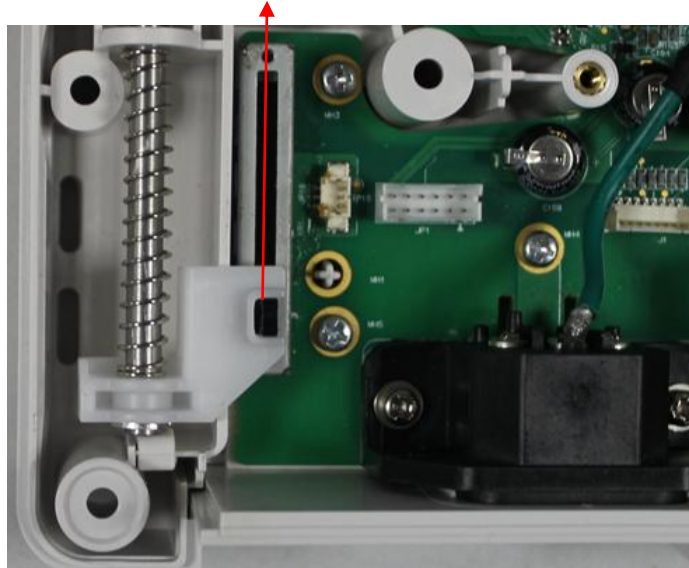


6. Remove the keypad board PCBA;
7. Remove the LCD (**Notes: replace LCD, the LCD should take the 4PCS sponge mat**)

5.3.8 Disassembly of main board PCBA

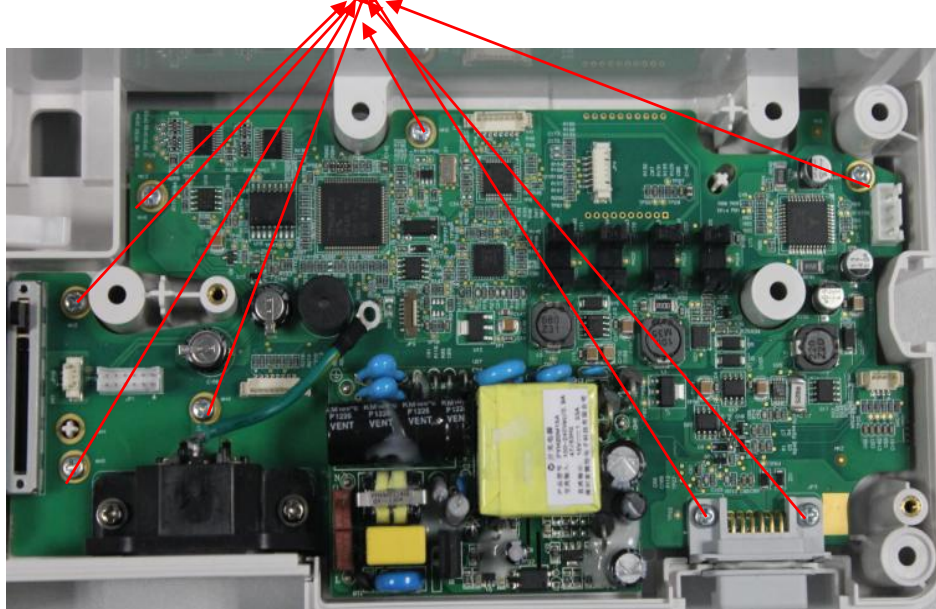
1. Remove the rear assembly firstly, then take out of main board PCBA.
2. Take short potentiometer fork, don't lock the handle of hort potentiometer, as shown below:

Take short potentiometer fork from here



3. Remove the 8 PT3×8 self-tapping screw, as shown below:

PT3×8 self-tapping screw



-
4. Remove 2 M3×12 double gasket screw, as shown below:

M3×12 double gasket screw



5. Remove the main board PCBA.

6 Parts

6.1 Description

This chapter lists the main components of syringe pump, including the Explosive view and parts material code, it's more convenient for maintenance personnel to identify the parts name when removal and replacement of parts.

The host structural system of syringe pumps as shown below:

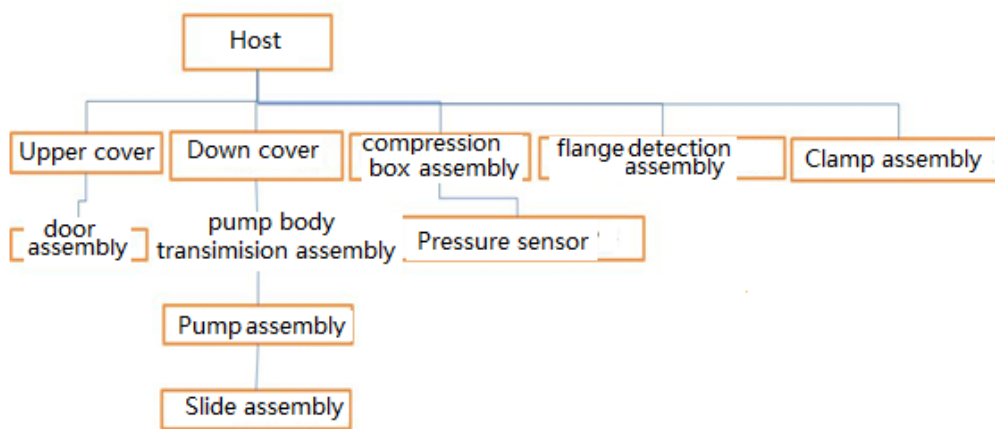


Fig. 1 Structural composition 1

6.2 Host

6.2.1 Explosive view

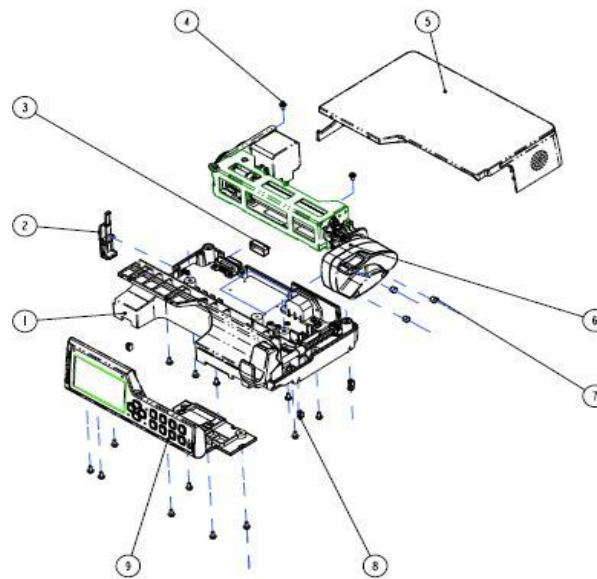


Fig. 2 Host explosive view

6.2.2 Parts list

NO.	Part Number	Part Name & Spec.	QTY
1	115-021570-00	SP5 rear enclosure assembly	1
2	043-004208-00	Multi-channel pumps hasp (common mode material)	1
3	043-004180-00	Multifunction interface cover	1
4	030-000305-00	Pan head screw M3x8(GB/T818) Nickel-plated steel double gasket	16
5	115-021569-00	SP5 UP cover assembly	1
6	115-021563-00	SP5 pump body assembly	1
7	049-000662-00	SP5 slider screw rubber plug	3
8	049-000664-00	Foot pad	4
9	115-021571-00	SP5 door assembly	1
10	050-001401-00	Wire FPC electron wire 10P separation distance 1.0mm length120mm	1
11	009-002714-00	Strap,3x150mm	2
12	009-004120-00	Wire electron wire 12P double head connection wire separation distance 2.0mm length100mm	1

Table 1 Host parts list

6.3 SP5 UP cover assembly

6.3.1 Explosive view

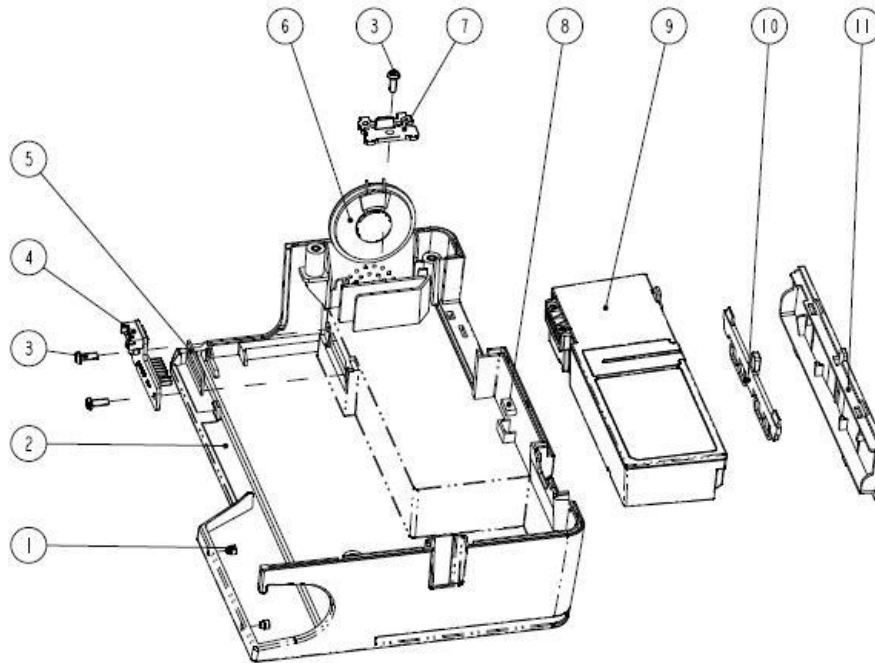


Fig. 3 UP cover explosive view

6.3.2 Parts list

NO.	Part Number	Part Name & Spec.	QTY
1	041-012461-00	SP5 door magnet	2
2	043-004188-00	SP5 UP cover	1
3	M04-003105---	pan head no tail tip cutting tapping screw PT3 × 8 green color zinc plated	3
4	051-001592-00	SP5 battery interface board PCBA	1
5	048-003783-00	MR pump battery adapter plate sponge	1
6	009-004119-00	Electron wire 3P2 wire loudspeaker assembly separation distance 1.25mm	1
7	043-004209-00	SP5 Speaker briquetting (common mode material)	1
8	048-003941-00	EVA cover(8.8×4.5×3)	1
9	022-000161-00	Lithium battery Li-ion 7.4V 4800mAh SK04B9003	1
10	043-004182-00	Battery box door clip	1
11	043-004183-00	SP5 battery box door	1

Table 2 UP cover parts list

6.4 SP5 Door Module

6.4.1 Explosive View

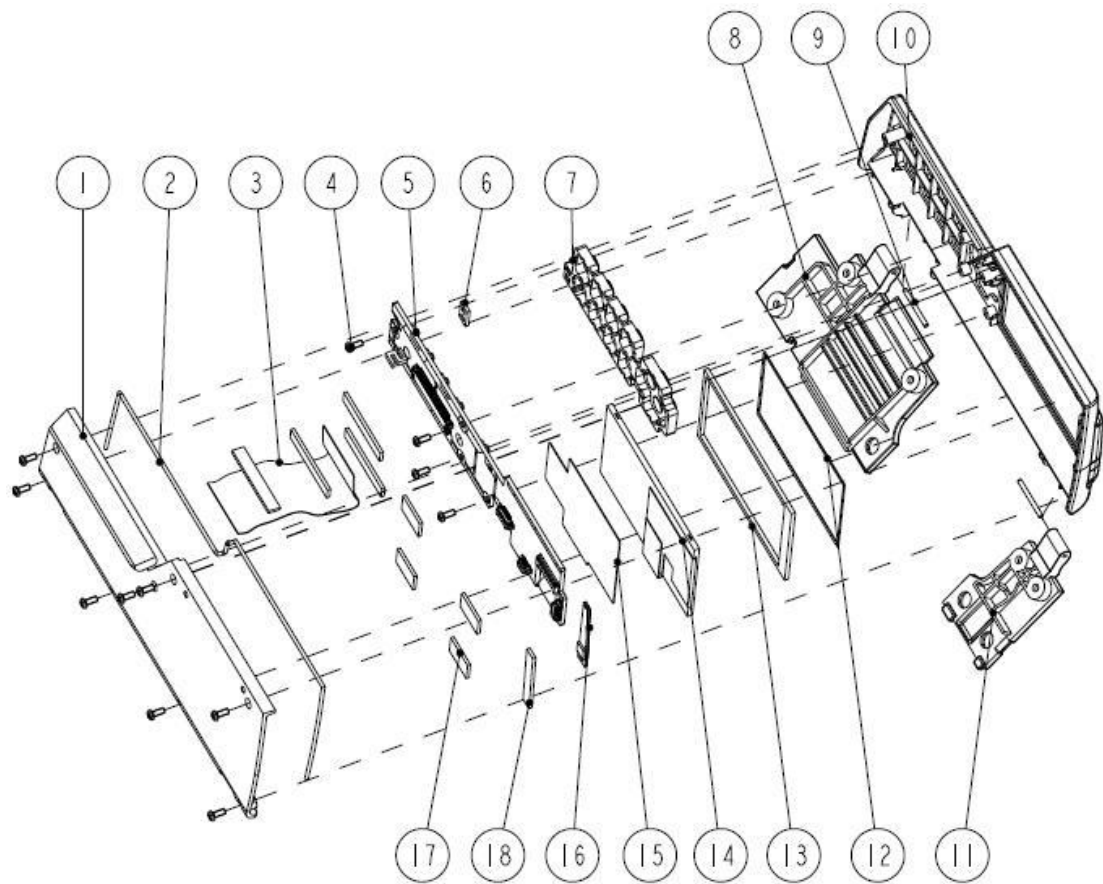


Figure 4 Door Explosive View

6.4.2 Parts List

NO.	Parts Number	Parts Name & Spec.	QTY
1	043-004186-00	SP5 Door back cover	1
2	M6G-020015---	Sebific duck,silicone rubber white 1.6mmODX0.8mmID	0.3
3	009-004212-00	Wire FFC electron beam 30P Gap1.0mm	1
4	030-000366-00	Cross PT2.6×6 Stainless Steel Dull Tapping Screw	12
5	051-001593-00	SP5 Keypad PCBA	1
6	048-003942-00	Keypad light-proof Covering	1
7	049-000672-00	Silica Gel Keypad	1
8	043-004217-00	SP5 Door's Right Spindle Cover	1
9	041-010088-00	SP5 Door Spindle	2
10	043-004185-00	SP5 Front Door Cover	1
11	043-004216-00	SP5 Door's Left Spindle Cover	1
12	047-012023-00	LCD Lens	1
13	048-003781-00	LCD Dust-proof Mat	1
14	021-000056-00	LCD Screen 3.5" 272*480 3.3V	1
15	047-012133-00	Insulating Strip	1
16	051-001591-00	SP5 Alarm Lamp PCBA	1
17	048-003002-00	Loudspeaker Mat	6
18	048-003782-00	SP5 Winding Displacement Tightening Mat	5

Table 3 Door Parts list

6.5 SP5 Rear Cover Module

6.5.1 Explosive View

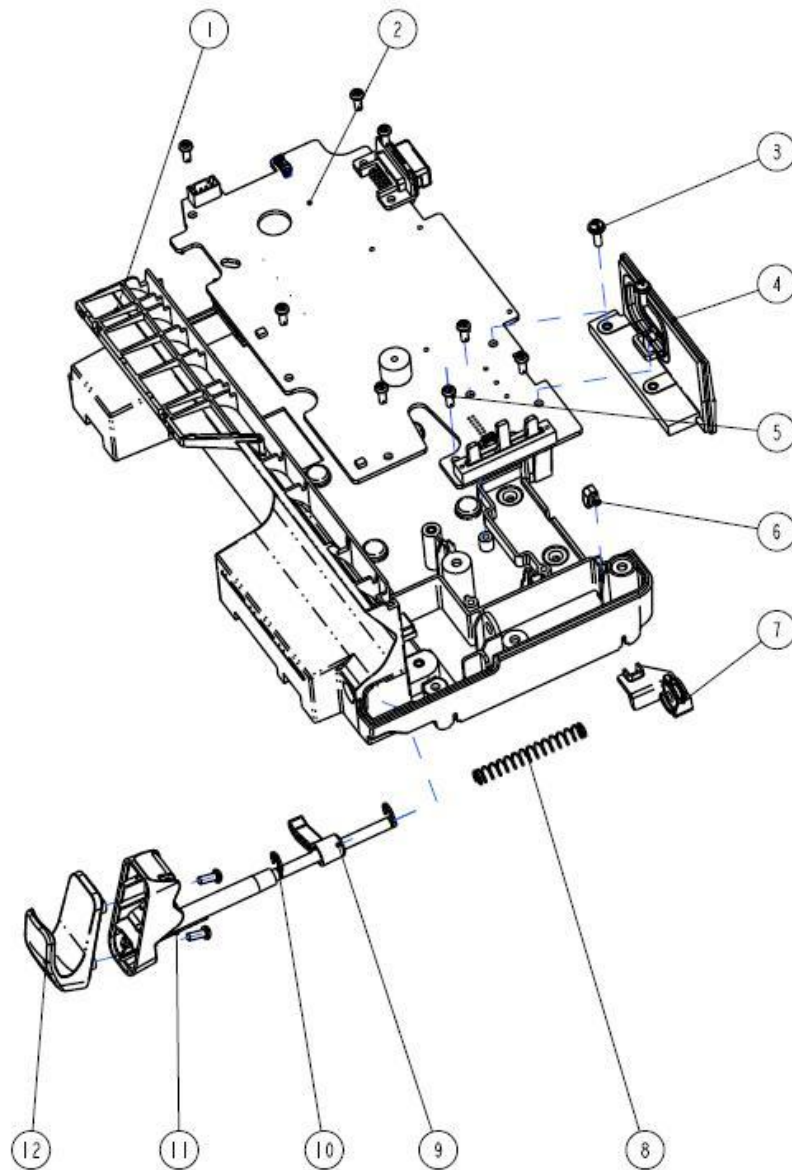


Figure 5 Rear Cover explosive view

6.5.2 Parts List

NO.	Parts Number	Parts Name & Spec.	QTY
1	043-004189-00	SP5 Bottom Casing	1
2	051-001582-00	SP5 Main Board PCBA	1
3	030-000242-00	M3x12 Cross Recessed Pan Head Screw Mat	2
4	043-004215-00	AC Socket Support	1
5	M04-003105---	Cross Tapping Screw PT3×8	10
6	049-000664-00	Foot Pad	1
7	043-004206-00	SP5 Short Potentiometer Shifting Fork	1
8	033-000309-00	Pull Rod Reset Spring (ID5.4×51)	1
9	043-004205-00	SP5Pull Handle Lever	1
10	030-000275-00	Split Washer $\varnothing$ 4×0.8 (GB/T896) Stainless Steel	2
11	043-004192-00	SP5 Pull Handle Back Casing	1
12	043-004191-00	SP5 Pull Handle Front Casing	1

Table 4 Rear Cover Parts

6.6 SP5 Pump Body Driven Module

6.6.1 Explosive View

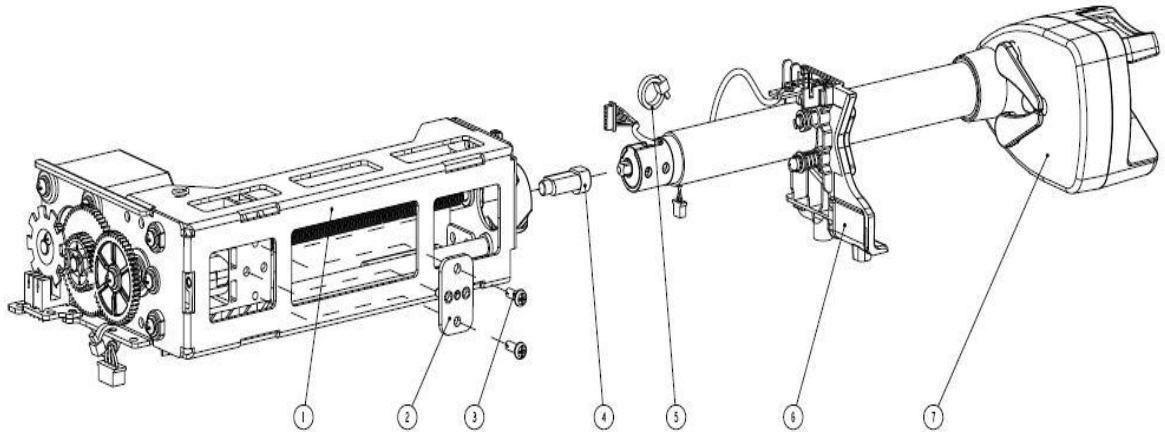


Figure 6 Pump Body Explosive View

6.6.2 Parts List

NO.	Parts No.	Parts Name & Spec.	QTY
1	115-021566-00	SP5 Pump Body Driven Parts	1
2	042-010543-00	SP5 Hollow Shaft Fixture Strip	1
3	M04-003105---	Cross Tapping Screw	2
4	041-010090-00	SP5 Open-Close Flexible Spindle	1
5	009-002714-00	Binding Wire 3*150mm	1
6	115-021567-00	SP5 Bulge Test Support Parts	1
7	115-021568-00	SP5 Push-Pull Box Parts	1

Table 5 Pump Body Parts

6.7 SP5 Sliding Module

6.7.1 Explosive View

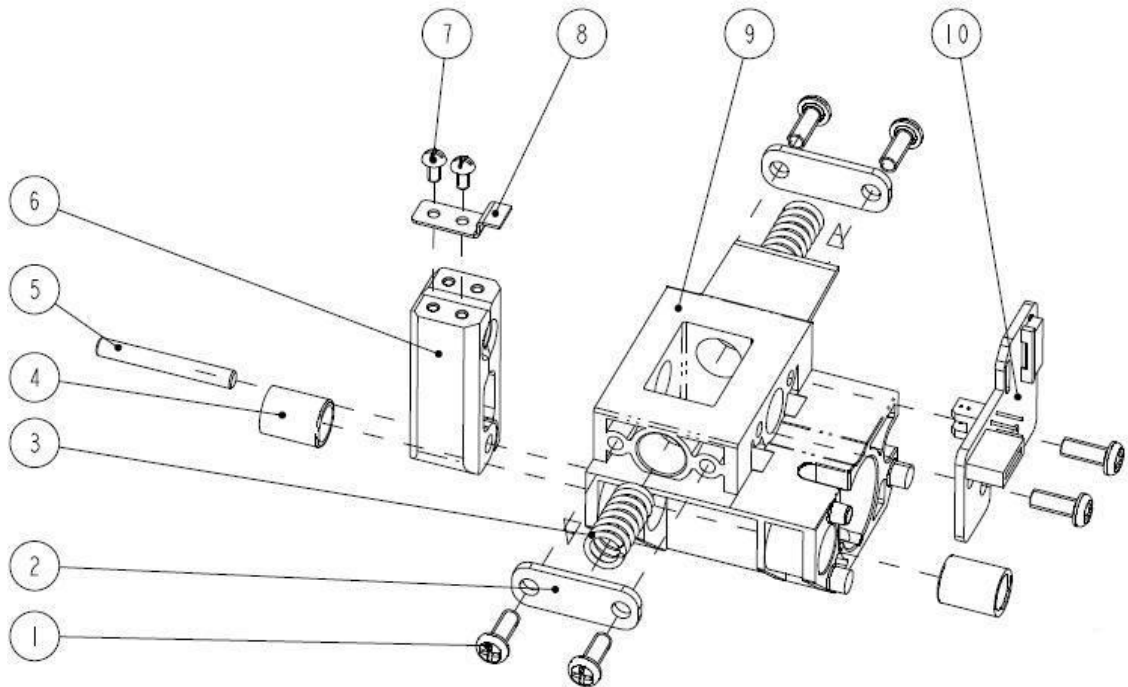


Figure 7 Sliding Module Explosive View

6.7.2 Parts List

No.	Parts No.	Parts Description & Spec.	QTY
1	M04-003105---	Cross Tapping Screw PT3×8	6
2	042-010546-00	SP5 Screw Nut Spring Press Board	2
3	033-000305-00	Screw Nut Spring Press	2
4	032-000355-00	Plastic Sliding lining JSM-0608-10	2
5	041-010092-00	Hinge Pin Φ3X23.5	1
6	041-010234-00	SP5 Screw Box	1
7	M04-051137---	Stainless Steel Cross Tapping Screw GB/T818-2000 M2X4	2
8	042-010647-00	SP5 Screw Nut Optical Coupling Shading Slice	1
9	043-004184-00	SP5 Sliding Module	1
10	051-001588-00	SP5 Screw Rod On-Off PCBA	1

Table 6 Sliding Module Parts List

6.8 SP5 Push-and-Pull Box Module

6.8.1 Explosive View

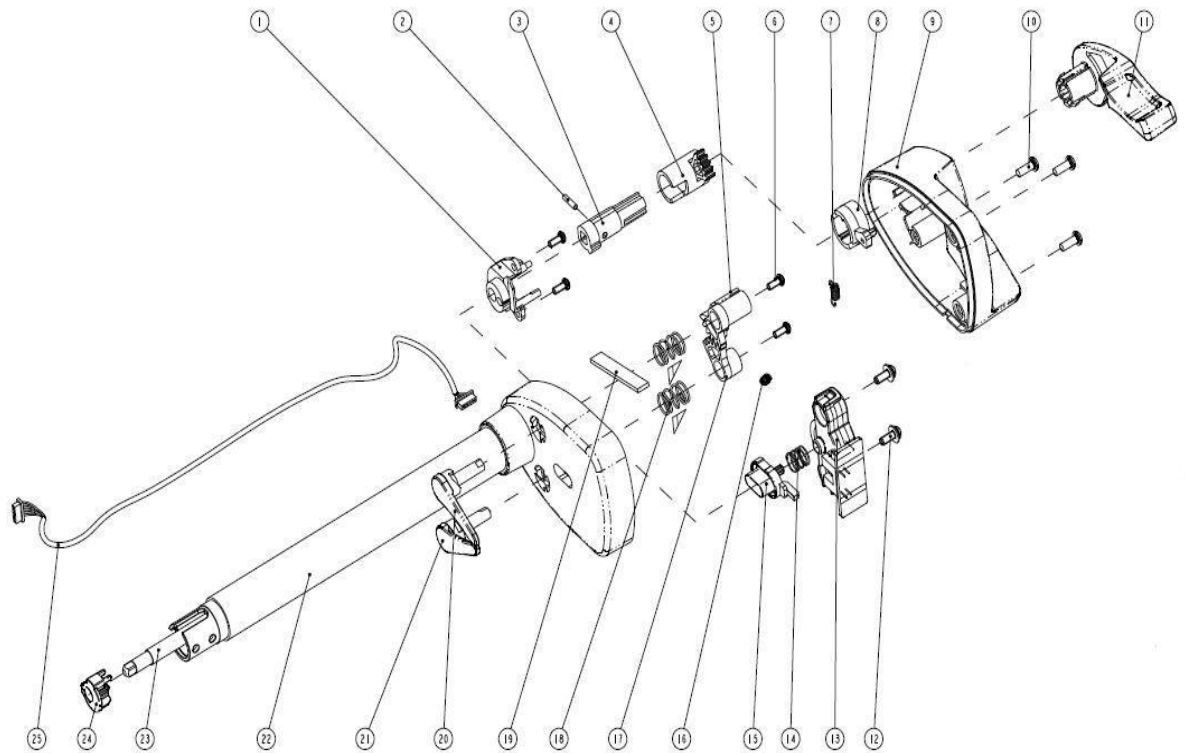


Figure 8 Push-and-Pull Box Explosive

6.8.2 Parts List

No.	Parts. No.	Parts Name & Spec.	QTY
1	043-004198-00	SP5 Twist Lever Fixture	1
2	041-010093-00	Hinge Pin $\Phi 2 \times L 8.6$	1
3	043-004178-00	SP5 Twist Lever Connector	1
4	043-004194-00	SP5 Twist Lever Gear	1
5	043-004195-00	SP5 Push-and-Pull Box Gear A	1
6	030-000366-00	Cross PT2.6×6 Stainless Steel Dull Tapping Screw.	4
7	033-000311-00	SP5 Hold Handle Spring(OD3.4×11.2)	1
8	043-004202-00	SP5 Hold Handle Fixture	1
9	043-004179-00	SP5 Push-and-Pull Box Back Casing	1
10	M04-003105---	Cross Tapping Screw PT3×8	3
11	043-004212-00	SP5 Push-and-Pull Box Hold Handle	1
12	030-000305-00	Cross Recessed Pan Head,M3×8(GB/T818)	2
13	115-022478-00	SP5 Pressure Sensor Parts	1
14	033-000307-00	Press Block Reset Spring (ID7.2×7.5)	1
15	043-004197-00	SP5 Push-and-Pull Box Press Button	1
16	033-000308-00	Clip Reset Spring(OD4.5×14.6)	1
17	043-004196-00	SP5 Push-and-Pull Box Gear B	1
18	033-000306-00	Clip Reset Spring(ID7.6×12)	2
19	048-003782-00	SP5 Winding Displacement Fixture Mat	1
20	043-004214-00	SP5 Right Clip	1
21	043-004213-00	SP5 Left Clip	1
22	043-004177-00	SP5 Push-and-Pull Box Front Cover (include Hollow Shaft)	1
23	041-012450-00	SP5 Twist Lever	1
24	043-004201-00	SP5 Twist Lever Guide Sleeve	1
25	009-004125-00	Electric Wire 6P double-end wire Spacing 1.25mm	1

Table 7 Push-and-Pull Box List

6.9 Pressure Sensor Module

6.9.1 Explosive View

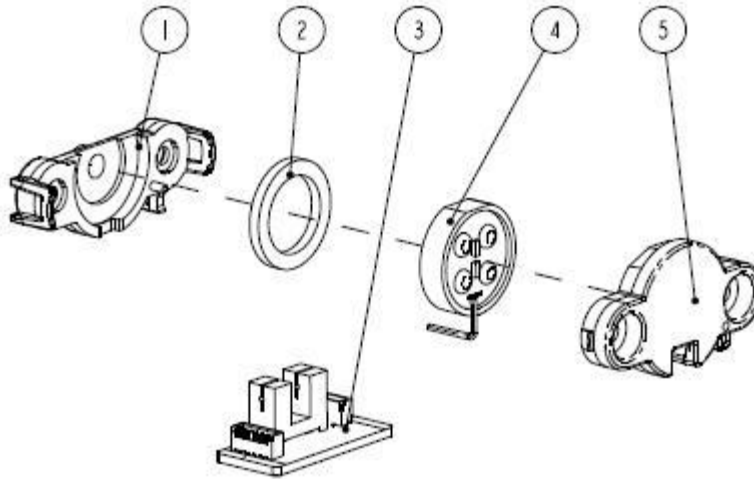


Figure 9 Pressure Sensor Module Explosive

6.9.2 Parts List

No.	Parts Number	Parts Name & Spec.	QTY
1	043-004199-00	SP5 Pressure Sensor Upper Cover	1
2	024-000424-00	Pressure Sensor Fixture Mat	1
3	048-003940-00	Pressure Sensor 25LBS 1.27mm Soldering Parts	1
4	043-004200-00	SP5 Pressure Sensor Bottom Cover	1
5	051-001587-00	SP5 Pressure Detection Board PCBA	1

Table 8 Pressure Sensor Module Parts List

6.10 SP5 Bulge Test Support Parts

6.10.1 Explosive View

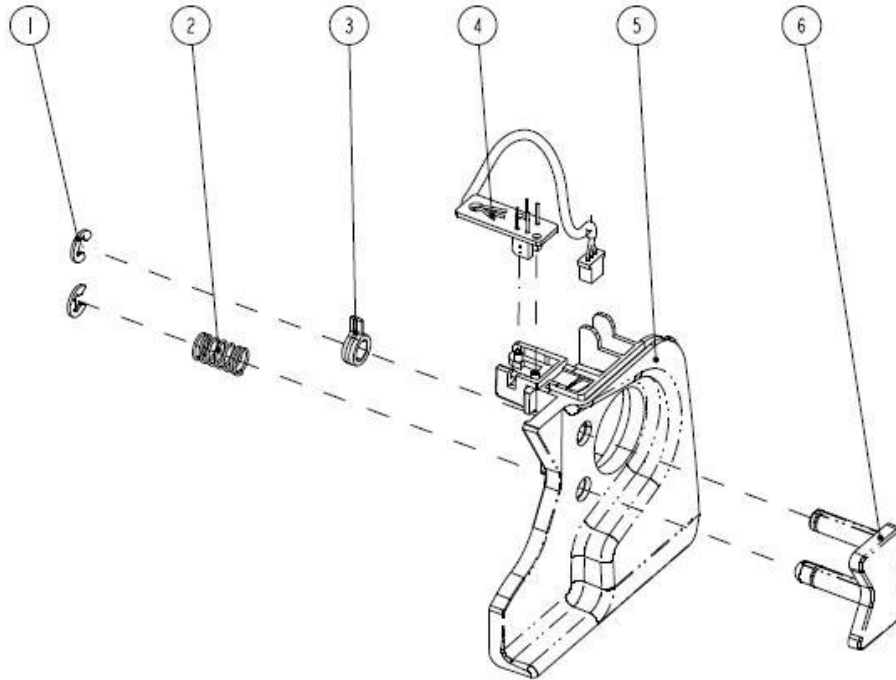


Figure 10 Bulge Test Support Explosive View

6.10.2 Parts List

NO.	Parts No.	Parts Name &Spec.	QTY
1	M04-021054---	Split Washer GB/T896-1986 3	2
2	033-000310-00	Bulge Fixture Spring	1
3	043-004207-00	SP5 Bulge Test Shading Slice	1
4	051-001589-00	SP5 Bulge Test Board PCBA	1
5	043-004190-00	SP5 Bulge Test Support	1
6	043-004211-00	SP5 Bulge Shift Fixture	1

Table 9 Bulge Test Support Parts List

6.11 Mindray Pump Clamp Parts

6.11.1 Explosive View

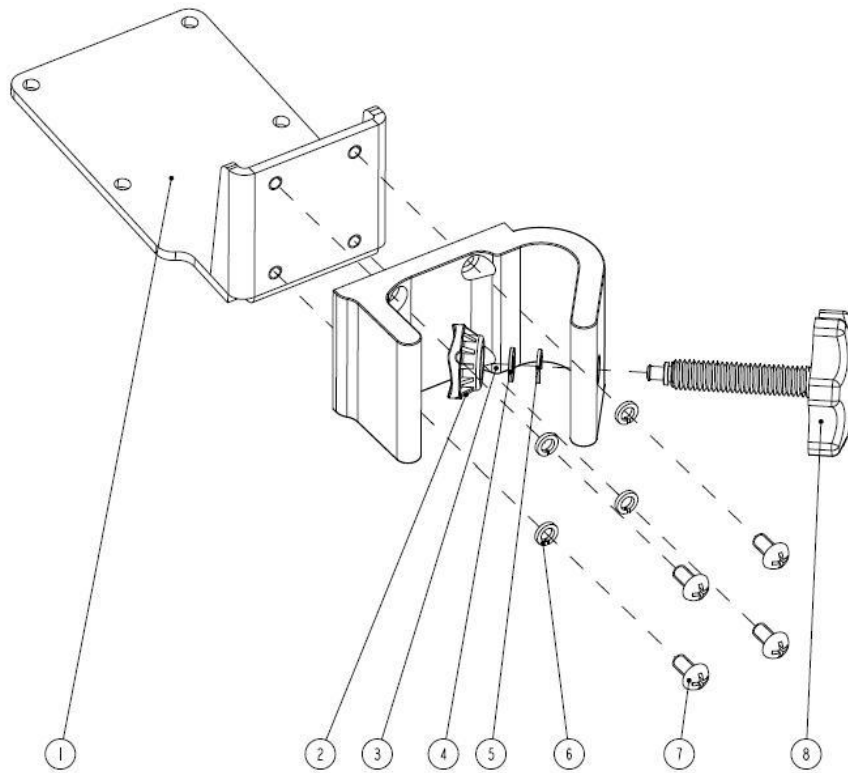


Figure 11 Mindray Pump Clamp Explosive View

6.11.2 Parts List

NO.	Parts No.	Parts Name &Spec.	QTY
1	042-010545-00	Fixture Clamp Connecting Board	1
2	043-003009-00	Fixture Clamp Plastic Upper Mat	1
3	045-000845-00	Fixture Clamp	1
4	M04-021073---	SUS316 flat washer-A GB/T97.1-2002 4 Dull	1
5	030-000275-00	Split Washer $\varnothing 4 \times 0.8$ (GB/T896) Stainless Steel	1
6	030-000266-00	$\varnothing 4$ Spring Mat	4
7	030-000219-00	cross recessed pan head screw, M4 $\times$ 8 (GB/T818)	4
8	043-003174-00	Fixture Clamp Knob	1

Table 10 Mindray Pump Clamp Parts List

7 Upgrading

7.1 Tools

During upgrade, you may need to use the following tools:

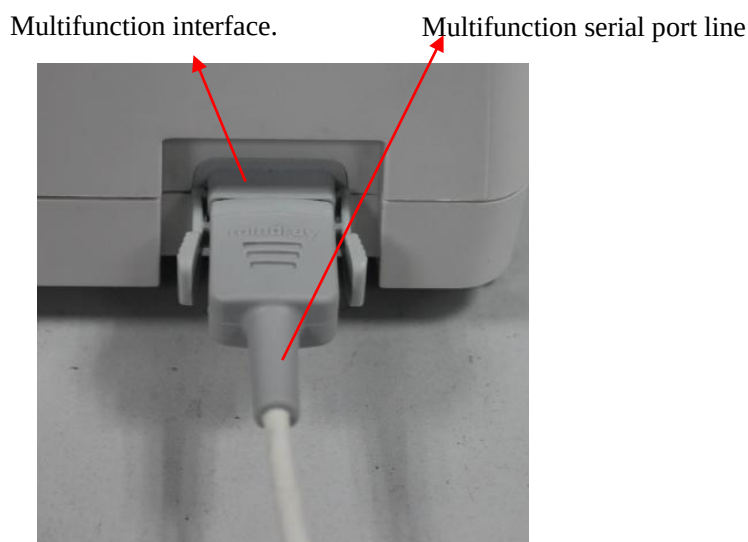
- ◆ Multifunction serial port line
- ◆ PC computer
- ◆ PC burning tool software

7.2 Software upgrade

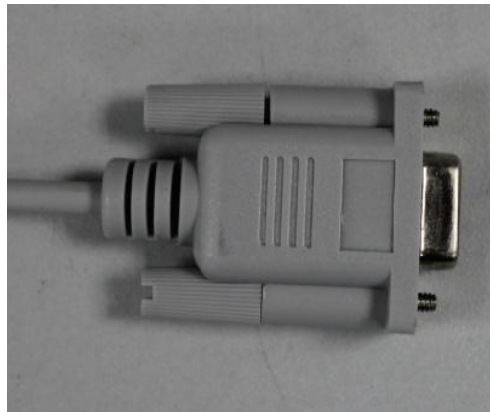
This syringe pump can support the software upgraded, the software burning can be operated directly with desk PC or mobile PC. By multifunction interface, this syringe pump can be connected to PC, the software can be upgraded, including main program, driver software, power board software, keypad board software, etc. (The upgrade of drug library, built-in syringe brad will be opened subsequently).

7.3 Software burning method

1. Insert the multifunction serial port line to multifunction interface of host, as shown below:



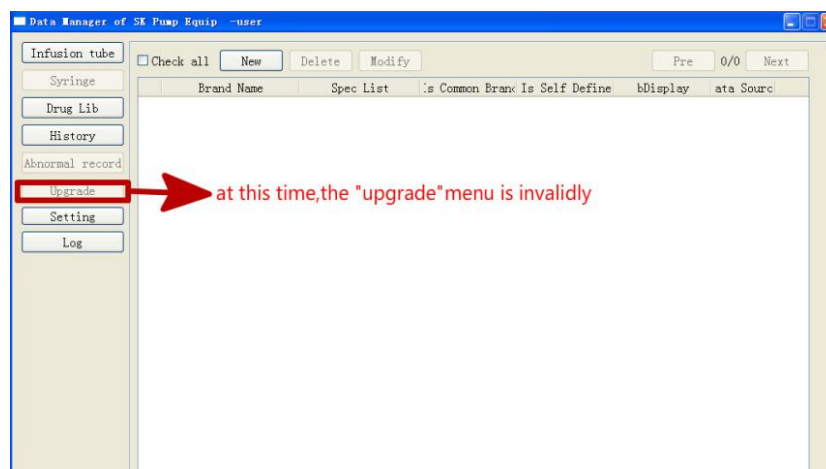
-
- The other side of multifunction serial port is standard BD9 serial port, as shown below, connect this side to computer.



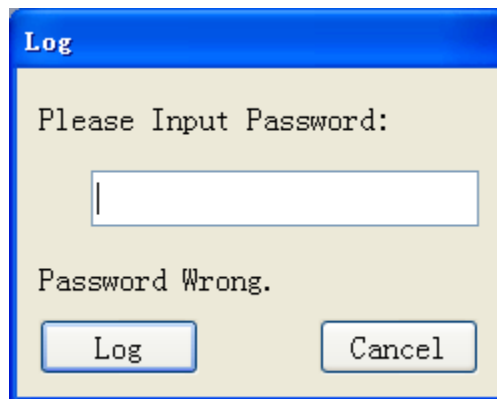
- No insert AC power line, open the battery box then remove the lithium battery.
- Press “start” key and “power” key together.
- After loosening the keys, install the battery, then the machine will enter boot, the BootLoader will be shown in display.



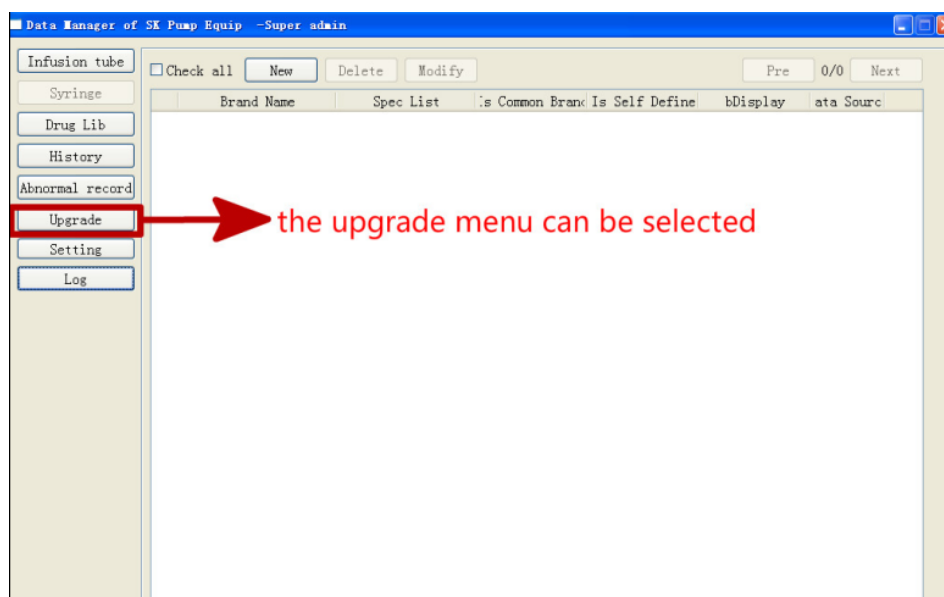
- Open the PC burning tool (G-GCRJ-0004), as shown below, at this time the “upgrade” menu is invalidly and can’t be selected.



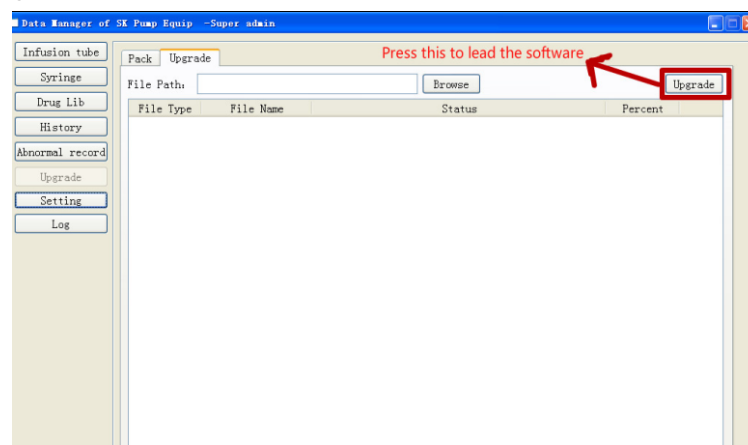
-
7. Click “administrator login”, the login password dialog box will be popped shown, as shown below:



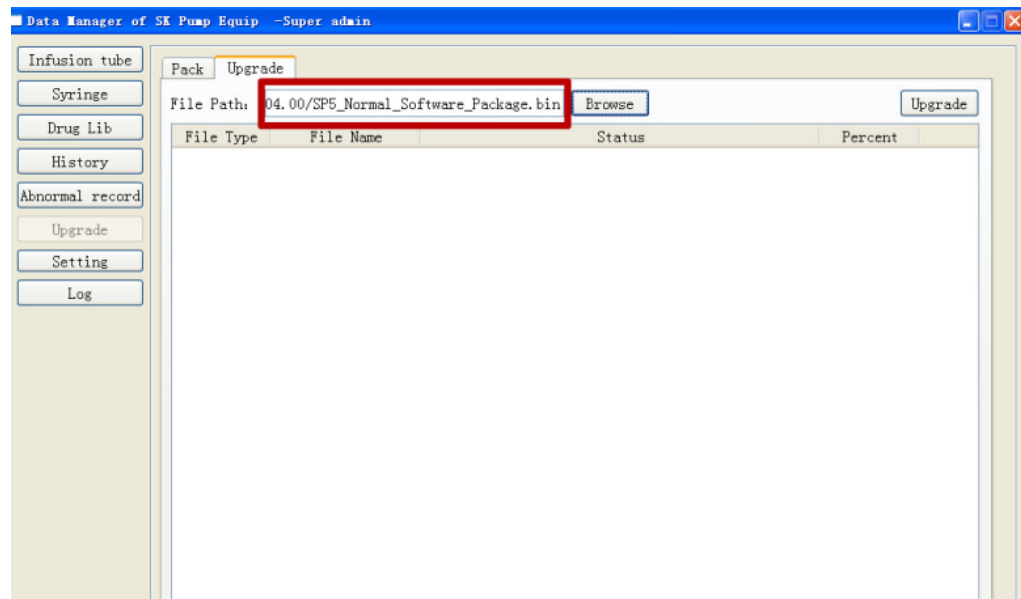
8. Enter password, right now the upgrade menu can be selected, as shown below:



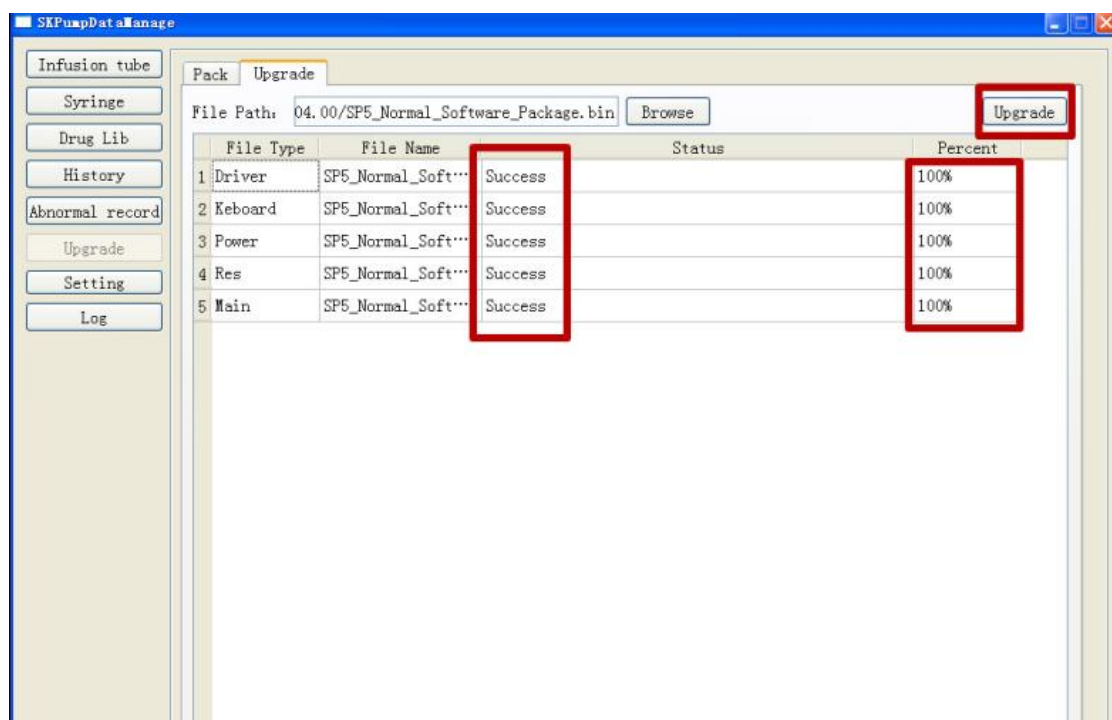
9. Choose “upgrade”, then show a new screen, as shown below;



10. In upgrade screen, choose “browse”, then choose the burning software box needed,, as shown below.



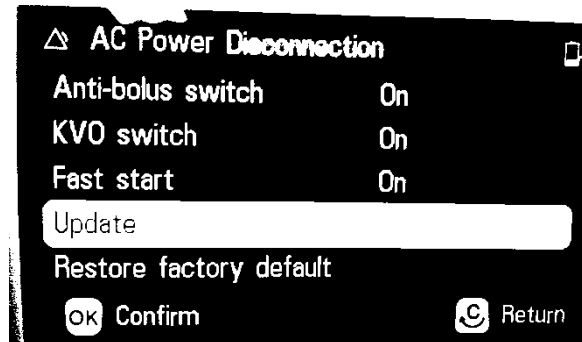
11. Click the “software” on the top right corner of PC burning tool software, software start burning.
12. After completed, the upgrade will display “successful”, the grade progress will show “100%”, as shown below:



13. Power off; disconnect the multifunction serial port line. The software upgrade is successfully finished.

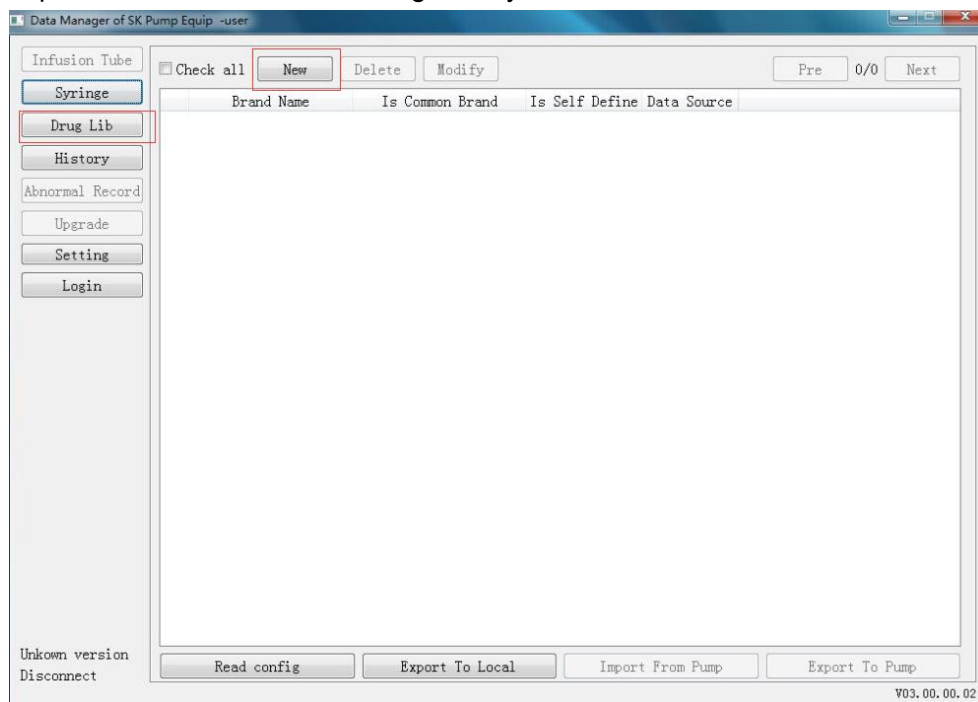
7.4 Update Drug Library

1. Turn on the pump ,enter the 'System Maintenance' interface.
2. Input password '6686',then press 'OK' confirm.
3. Choose 'Update'.
4. Connect PC and Pump with Multi-functional serial port.



7.4.1 One kind drug download

1. Open PC software ,choose 'Drug Library' ,then click 'New'



2. Write the drug information, then confirm.

new

Basic information

*Drug Name Mindray

*Drug code 001X

Conc.

Drug Vol. 0.00 ng Liquid Vol. 0.00 ml Conc. 0.00 ng/ml

Rate limit

☒ Rate ☐ Dose

Min 0.1 ml/h Max 0.1 ml/h Default 0.1 ml/h

Bolus limit

☒ Rate ☐ Dose

Min 0.1 ml/h Max 0.1 ml/h Default 0.1 ml/h

OK Cancel

3. Export the drug to pump.

Infusion Tube

Syringe

Drug Lib

History

Abnormal Record

Upgrade

Setting

Login

Connected

Check all New Delete Modify

Pre 1/1 Next

	Drug code	Drug Name	Data Source	Conc.	Min rate/dose	Max rate/dose	efault rate/dos	Bo
1	☒ 001	Mindray	User Add ...	0 ng/ml	0.1 ml/h	0.1 ml/h	0.1 ml/h	0.1

choose the code

Warning

Data unable to recover.

Yes No

confirm

export to pump

Read config Export To Local Import From Pump Export To Pump

V03.00.00.02

7.4.2 Drug library batch update

1. Prepare for drug library source file , 'xml' format.
2. Click on 'Read configuration file' ,then choose the target drug library ,then export to pump.

The screenshot shows a software interface for managing a drug library. On the left is a sidebar with buttons: Infusion Tube, Syringe, Drug Lib, History, Abnormal Record, Upgrade, Setting, and Login. The main area contains a table with columns: Drug code, Drug Name, Data Source, Conc., Min rate/dose, Max rate/dose, and default rate/dose. The table lists 18 drugs, all with 'Local' as the data source and '0' for the rates. Above the table are buttons for 'Check all', 'New', 'Delete', and 'Modify'. Below the table are buttons for 'Read config', 'Export To Local', 'Import From Pump', and 'Export To Pump'. The 'Read config' and 'Export To Pump' buttons are highlighted with red boxes. At the bottom right, the text 'V03.00.00.02' is visible.

Drug code	Drug Name	Data Source	Conc.	Min rate/dose	Max rate/dose	default rate/dose
1	Metoprolol	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
2	Amiodarone	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
3	Esmolol	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
4	Lidocaine	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
5	Propafenone	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
6	Nitroglyc...	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
7	Nicardipine	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
8	Isosorbid...	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
9	Diltiazem	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
10	Digoxin	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
11	Dobutamine	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
12	Dopamine	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
13	Epinephrine	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
14	Isoprenal...	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
15	Norepinep...	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
16	Atropin	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
17	Succinylc...	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h
18	Pancuronium	Local	0 ng/ml	0 ml/h	0 ml/h	0 ml/h

3. System tooltip shown as below:

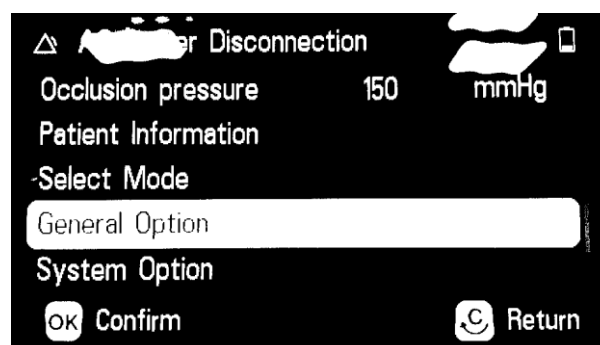
This screenshot is similar to the previous one, but it includes a 'Warning' dialog box in the center. The dialog box has a yellow warning icon and the text 'Data unable to recover.' with 'Yes' and 'No' buttons. The 'Yes' button is highlighted with a red box. The background table and interface elements are the same as in the previous screenshot.

-
4. Downloading success.

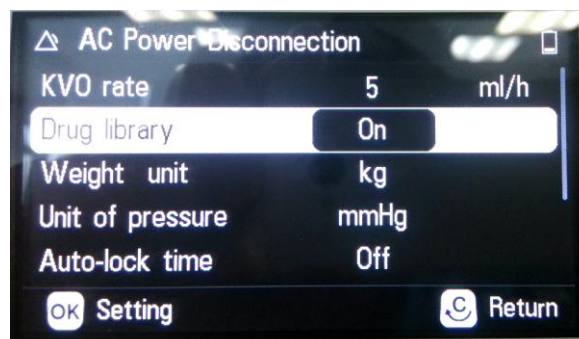


7.4.3 Check the downloading drug

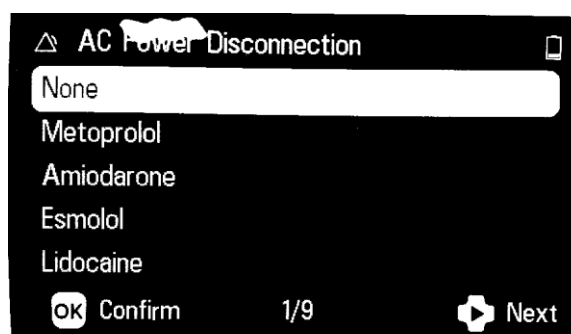
1. Firstly ,enter the general option interface.



2. Then turn the drug library option to on.



3. Back to the main menu ,check the drug downloading.



Note: click on "save configuration" also can achieve target, choose drug item needed to save to PC.

7.5 Add Syringe Brand

1. Connecting SP5 and PC with multifunction series ports cable (as per 7.4 update drug library method).
2. After connection ,choose 'syringe ' option.

Connected

3. Click 'New', add the syringe brand information.

	Brand Name	Spec	Full Length	Outside Dia.	Empty Pos	Syring Cali. Time	Press Factor	re
1	Mindray	5ml				2013年1月1日 0:...		
2	Mindray	10ml				2013年1月1日 0:...		
3	Mindray	20ml				2013年1月1日 0:...		
4	Mindray	30ml				2013年1月1日 0:...		
5	Mindray	50ml				2013年1月1日 0:...		

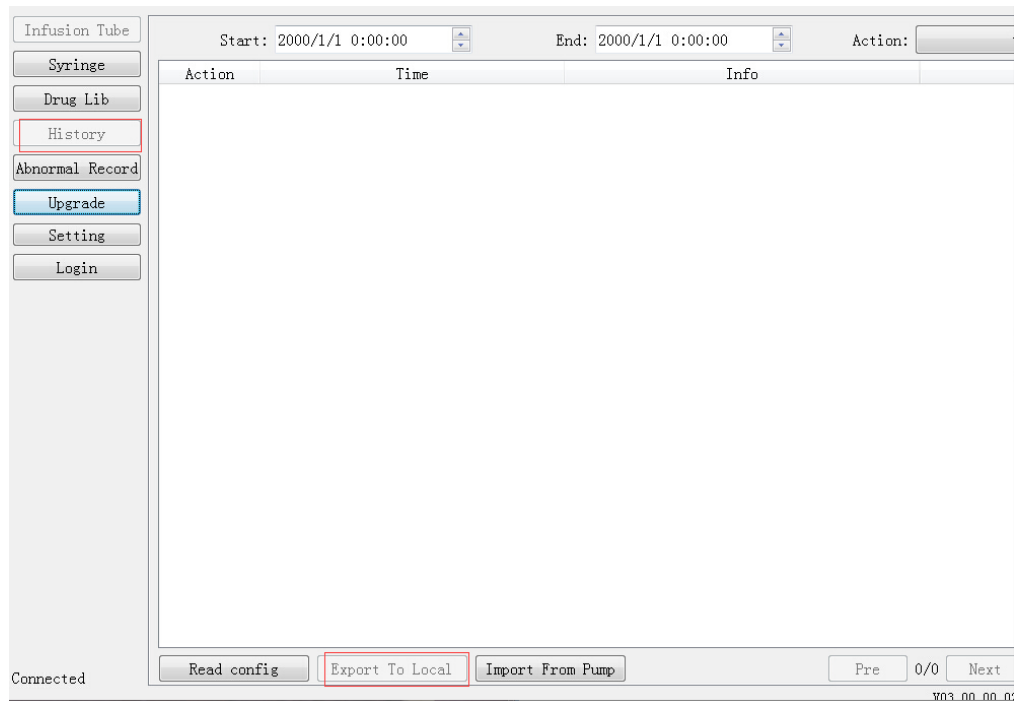
4. Export all syringe brand ,choose the brand added.

	Brand Name	Spec List	Is Common Brand	Is Self Define	Data Source
1	☒ Mindray	5ml, 10ml, 20ml, 3...	☐ Yes ☒ No	☐ Yes ☐ No	User Add ...
2	☐ B. Braun		☐ Yes ☒ No	☐ Yes ☐ No	Pump
3	☐ B-D		☒ Yes ☐ No	☐ Yes ☐ No	Pump
4	☐ Shuangge		☐ Yes ☒ No	☐ Yes ☐ No	Pump
5	☐ Jierui		☒ Yes ☐ No	☐ Yes ☐ No	Pump
6	☐ Terumo		☐ Yes ☒ No	☐ Yes ☐ No	Pump
7	☐ SuYun		☐ Yes ☒ No	☐ Yes ☐ No	Pump
8	☐ ShanChuan		☐ Yes ☒ No	☐ Yes ☐ No	Pump
9	☐ QiaoPai		☐ Yes ☒ No	☐ Yes ☐ No	Pump
10	☐ HuaFu		☐ Yes ☒ No	☐ Yes ☐ No	Pump
11	☐ Brand1		☐ Yes ☒ No	☐ Yes ☐ No	Pump
12	☐ Brand2		☐ Yes ☒ No	☐ Yes ☐ No	Pump

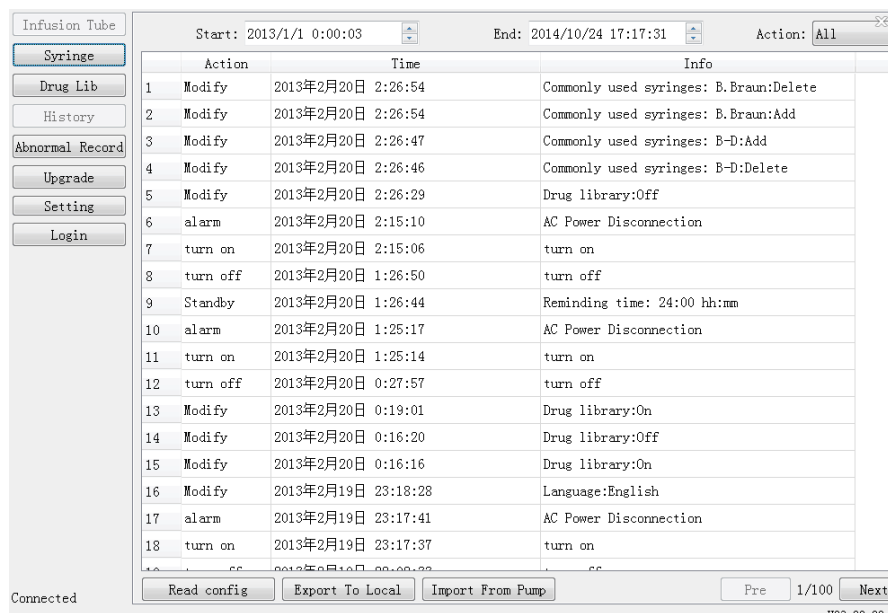
5. Restart the pump ,confirm the brand succeed in adding to the pump or not
Note: also click 'save to configuration file' , and choose the syringe brand, specification under the PC tool toolbar.

7.6 Export History Record

1. Connecting SP5 and PC with multifunction series ports cable (as per 7.4 update drug library method).
2. After connected, choose “History Record”, Click “Export History”.
3. PC automatically read the history record.

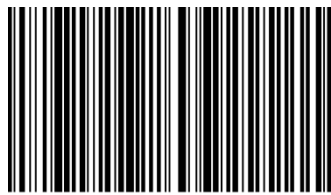


4. Download the file and save.



5. Check and confirm the file .

FOR YOUR NOTES



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